MODEL SET 11

लोक सेवा आयोग

नेपाल इन्जिनियरिङ सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह,

राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

समय: ३ घण्टा **पूर्णाङ्क: १००**

पत्र: द्वितीय **विषय: Technical Subject**

तलका प्रश्नहरुको उत्तर Section अनुसार छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ।

Section A

1. **Describe under reinforced, over reinforced and balanced RCC section.[5]**

- **Balanced Section**
  - → It is such type of section (theoretical condition) where the amount of steel provided is "just right" as comparison to the concrete.
  - → Failure of section occurs as simultaneous failure i.e., both the concrete reaches its ultimate crushing strain, and the steel reaches its yield strain at the exact same time.
  - → It is also known as critical or economical section.
- **Under Reinforced Section**
  - → In an under-reinforced section, the amount of steel provided is **less** than what is required for a balanced section.
  - → Failure of section occurs as ductile failure i.e., steel reaches its yield strength before the concrete reaches its maximum compressive strain.
- **Over Reinforced Section**
  - → In an over-reinforced section, the amount of steel provided is more than what is required for a balanced section.
  - → Failure of section occurs as brittle failure i.e., concrete reaches its ultimate compressive strain (crushing) before the steel yields.

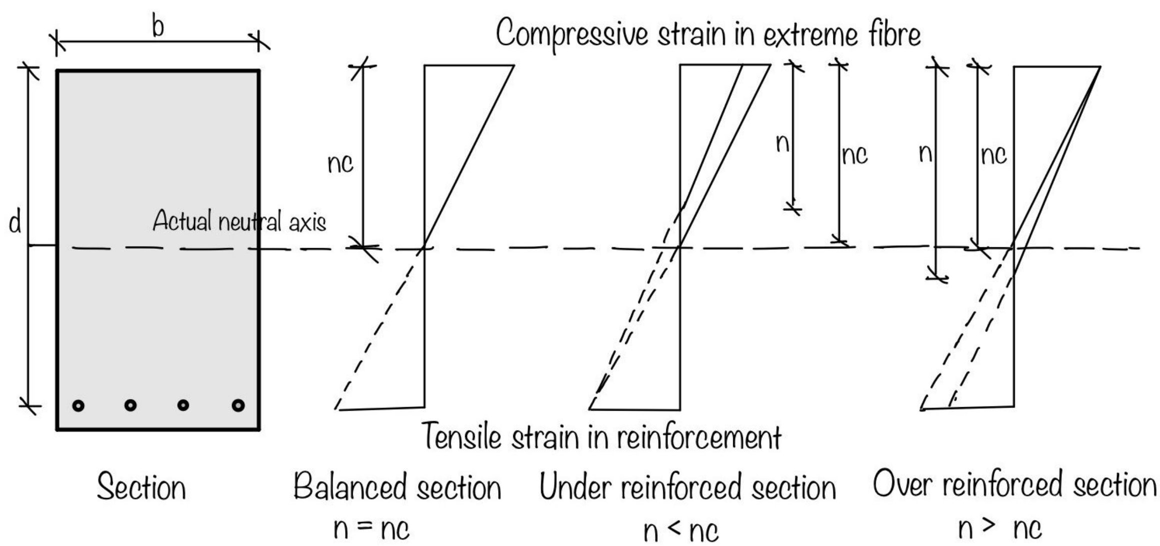


Figure 1: Different types of RCC Beam sections

2. **Design a combined footing for two columns 32 cmx 32 cm carrying load of 60,000 kg and 40 cm carrying load of 80,000 kg. The columns are spaced at 3.4 m centers, and the hearing capacity of soil is 15 tones/m² Calculate the depth of the footing, reinforcement required at transverse section and cantilevers. [10]**

Step 1: Base Area for footing $= \dfrac{1.1 \times (600 + 800)}{150} = 10.26\ \text{m}^2$

(considering extra 10% for self wt. of soil above footing)

Step 2: Calculating line of action of resultant of column load measured from exterior column

$$Q \times \bar{x} = Q_2 \times x_2$$

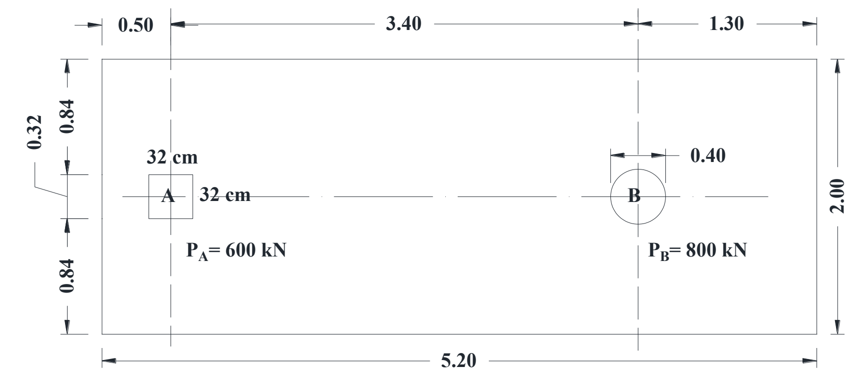
$$\bar{x} = \frac{800 \times 3.4}{1400} = 1.94\ \text{m}$$

Step 3: Calculate Length and Width of Footing

Let us provide a projection of 0.5 m from the center of exterior column

Then, Length of Footing = 2 (0.5+0.32/2+1.94) = 5.2 m

Width of Footing = 10.26/5.2 = 1.97 m (Adopt B=2 m)



Step 4: Calculate Longitudinal BM and SF

Net upward soil pressure $= \dfrac{1.5 \times (600 + 800)}{5.2 \times 2} = 201.92\ \text{kN/m}^2$

Net upward soil pressure per unit length $= 201.92\ \text{kN/m}^2 \times 2 = 403.84 \approx 405\ \text{kN/m}$

Maximum SF at centerline of column A

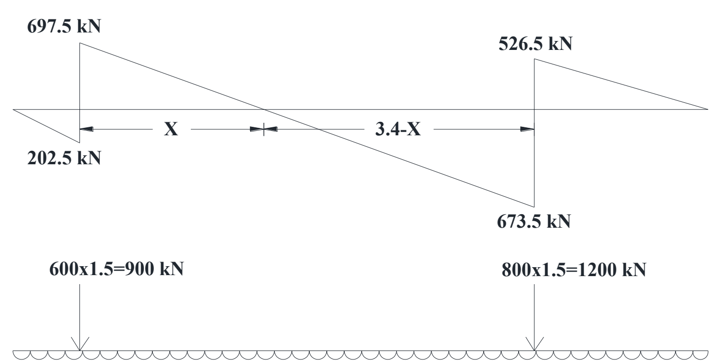
$v_1 = -405 \times 0.5 = -202.5\ \text{kN}$

$v_2 = -202.5 + 900 = 697.5\ \text{kN}$

Maximum SF at centerline of column B

$v_1 = 405 \times 1.3 = 526.5\ \text{kN}$

$v_2 = 526.5 - 1200 = 673.5\ \text{kN}$



Point of zero SF from CL of column A,

$\dfrac{697.5}{x} = \dfrac{673.5}{3.4 - x} \gg x = 1.73\text{m}$

Maximum BM computed from left side,

$= 405 \times \dfrac{(0.5 + 1.73)^2}{2} - 900 \times 1.73 = -549.987\ \text{kNm}$

Maximum BM computed from Right side,

$= 405 \times \dfrac{(1.3 + 1.67)^2}{2} - 1200 \times 1.67 = -217.76\ \text{kNm}$

Take maximum moment for computing depth of footing (considering Fe 415 grade steel and M25 Concrete)

$$549.987 \times 10^6 = 0.36\frac{X_u}{d}\left(1 - 0.42\frac{X_u}{d}\right)bd^2 f_{ck} = 0.36 \times 0.48(1 - 0.42 \times 0.48) \times 1000 \times d^2 \times 25$$

$\therefore$ d = 399.3 mm ≈ 400 mm
Adopt overall depth = 800 mm and effective depth as 750 mm
Main Longitudinal reinforcement:

$$M_u = 0.87 f_y A_{st} d \left(1 - \frac{A_{st} f_y}{f_{ck} bd}\right)$$

$$549.987 \times 10^6 = 0.87 \times 415 \times A_{st} \times 750 \left(1 - \frac{415 A_{st}}{25 \times 1000 \times 750}\right)$$

Solving:

$$A_{st} = 2250 \text{ mm}^2$$

Provide (practical): 20 mm bars @ 125 mm c/c

$$A_{prov} = 2510 \text{ mm}^2/\text{m}$$

Transverse Reinforcement

$$A_{st(min)} = 0.12\% \times b \times D$$
$$= 0.0012 \times 2200 \times 800 = 2112 \text{ mm}^2$$

Provide: **20 mm bars @ 140 mm c/c**

$$\left(A_{st(provided)} \approx \pi \times \frac{20^2 \times 1000}{140} \text{mm}^2 \approx 2244 \text{ mm}^2\right)$$

Cantilever Reinforcement
For projection **1.3 m**:
Moment:

$$M = \frac{q_u l^2}{2} = \frac{405 \times 1.3^2}{2} = 342.22 \text{ kNm}$$

Required steel:

$$A_s \approx 1300 \text{ mm}^2$$

Provide: **16 mm bars @ 125 mm c/c** (top, cantilever zone)

3. **Write short notes on methods and equipment of soil compaction in civil infrastructures construction. [5]**

Soil compaction is the mechanical process of increasing the density of soil by reducing air voids, that tends to increase bearing capacity, reduce settlement, and decrease permeability in civil infrastructure.

Field Methods of Soil Compaction & Their Suitability

It depends primarily on the **area** and the **type of soil**.

→ Manual and Small-Scale Methods

These methods are best suited for **narrow areas** where heavy machinery cannot maneuver.

- **Rammer (Hand Rammer):** A manual tool used for localized compaction.
- **Vibratory Plate Jumper:** A mechanical version of a rammer with a handle for better control.

Best for: Inside buildings, Behind retaining walls**,** Trenches and so on.

→ Large-Scale Methods (Rollers)

For **widely spread areas** such as roads, runways, and dams, different types of rollers are used based on the soil composition.

General Applications for Large Equipment

Roller Type	Ideal For	Specific Notes
Steel Drum Roller	Asphalt & Coarse-grained soil	Used for surface dressing and smooth finishing.

Pneumatic Tyred Roller	Coarse-grained soil	Best when fines (silt/clay) are **less than 20%**.
Sheep Foot Roller	Cohesive Soil	Features "feet" that penetrate the soil; best when fines are **greater than 20%**.
Vibrating Roller	Sand	Uses vibrations to rearrange particles into a dense state.
Grid Roller	Gravel & Weathered Rock	Has a grid-like surface with holes to break up and compact stony material.

4. **Discuss the causes of slope failures. Explain the concept of bioengineering and its advantages and disadvantages in slope stabilization in Nepal's challenging terrains. [3+7]**

Causes of Slope Failures

Slope failure occurs when the driving forces (primarily gravity) exceed the resisting forces (shear strength) of the soil or rock mass. The causes of slope failure can be categorized into factors that **increase shear stress** (driving forces) and factors that **decrease shear strength** (resisting forces).

1. Factors Increasing Shear Stress (Driving Forces)

These factors increase the force by pulling the material down the slope:

- **External Loading:** Manmade structures at top of slope, increases the downward gravitational force.
- **Increased Unit Weight:** When soil becomes saturated with rainwater, increases the total weight of the soil mass.
- **Steeping of the Slope:** Human activities (like excavation at the "toe" or base of the slope) or natural erosion can make a slope steeper, increasing the component of gravity acting parallel to the slope.

2. Factors Decreasing Shear Strength (Resisting Forces)

These factors weaken the soil or rock's ability to "hold on" to the slope:

- **Weathering:** Weathering break down the mineral bonds in rock and soil, tends to lower shear strength.
- **Geological Discontinuities:** Pre-existing weak planes, such as joints, faults, or bedding planes at the slope face, provides a natural path for failure to occur.
- **Loss of Vegetation:** Roots act as "mechanical anchors" that bind soil together. Cutting trees or plants (through deforestation or fire) removes this reinforcement.

Bioengineering is the use of living plant materials, often in combination with small-scale civil engineering structures, to stabilize slopes and control erosion.

Advantages in Nepal's Terrain

- **Cost-Effectiveness:** It costs significantly less than conventional reinforced concrete structures because it uses local materials and labor.
- **Environmental Sustainability:**

→ It restores ecosystems, increases biodiversity, and reduces surface runoff by absorbing rainwater.

→ Roots provide mechanical reinforcement by binding soil particles with shallow soil layers to deeper bedrock.

→ Leaves and branches intercept rainfall, reducing the impact of rain drop on the soil surface to reduce soil erosion.

- **Adaptability:** Unlike rigid structures, vegetative systems are flexible and can absorb minor ground movements during movement of vehicles.

Disadvantages and Challenges

- **Time-Dependent:** Vegetation requires time (often 2–3 years) to establish a root system strong enough to stabilize the soil.

- **Less Control for Deep Failures:** It is generally effective for shallow-seated landslides but cannot be workable for deep-seated failures or critical infrastructure loads.
- **Maintenance Requirements:** Require intensive monitoring and maintenance by local communities during the early stages of establishment. (should be protected from animal grazing)

Section B

5. Describe the procedures laid down in the guidelines on the implementation and management of water supply project, 2047 for the formation of water user's committee, also mention its duties and responsibilities.[10]

The **Guidelines on the Implementation and Management of Water Supply Project (Directives), 2047** (1990) established a participatory framework for rural water supply in Nepal. These guidelines shifted the focus from government-led construction to community-based management through the formation of **Water User's Committees (WUC)**.

Procedures for Formation of WUC

Under the 2047 directives, the formation process follows a participatory approach designed to ensure local ownership:

→ **Project Identification:** Schemes are identified based on community demand or local needs through a preparatory phase.

→ **User Orientation:** Local residents (beneficiaries) are oriented on the project's technical aspects, costs, and their long-term management roles.

→ **Mass Meeting:** A general assembly of all potential water users is convened to discuss the project and formally decide on committee formation.

→ **Democratic Selection:** The committee members are selected or elected by the users during the mass meeting. Current standards (evolving from these early guidelines) typically aim for **30–40% female representation** and inclusion of marginalized groups.

→ **Formal Registration:** Once formed, the WUC must be registered with the relevant government authority (often the District Water Resources Committee) to gain legal status for tapping water sources and opening bank accounts.

Duties and Responsibilities

The primary role of the WUC is to act as the legal and operational representative of the beneficiaries. Their core responsibilities include:

→ **Project Planning and Design:** Actively participating in the survey, technical layout, and approval of the water supply system.

→ **Construction Management:** Mobilizing local labor, managing construction materials, and overseeing the installation process (e.g., pipe laying and reservoir building).

→ **Financial Management:** Opening and maintaining a project bank account, collecting **water tariffs (fees)** from households, and managing funds for regular operations and emergency repairs.

→ **Operation and Maintenance (O&M):** Ensuring the continuous functionality of the system, sourcing village maintenance workers (VMWs), and performing regular infrastructure monitoring.

→ **Resource Allocation and Conflict Resolution:** Managing the equitable distribution of water among users and resolving disputes related to water use or system maintenance.

→ **Water Quality Assurance:** Implementing Water Safety Plans (WSP) and conducting regular tests (e.g., using P/A vials or chlorination) to ensure safe drinking water.

6. The following stream flow records are obtained from a gauging station.

Time (Hr.)	0	12	24	36	48	60	72	84	96
Q (cumec)	5	8	20	50	30	20	10	6	5

Determine volume of flood run off, base flow, surface run off and peak flood.

By looking at the table, the maximum value for Q occurs at T = 36 hours. **Peak Flood = 50 cumecs**
Base Flow is assumed to be constant, represented by the initial/final flow. **Base Flow Q_b = 5 cumecs**
Calculate Surface Runoff (Direct Runoff)
Surface runoff is the total flow minus the base flow for each time interval.

Time (Hr)	Total Flow (Q)	Direct Runoff (Q_dr=Q−Qb)
0	5	0
12	8	3
24	20	15
36	50	45
48	30	25
60	20	15
72	10	5
84	6	1
96	5	0
Sum		**104 cumecs**

Calculate Volume of Flood Runoff
Total Volume of Flow (Flood Runoff)

$$V_{total} = \Delta t \times \left(\sum Q\right)$$

$$\sum Q = \left[\frac{5+5}{2} + 8 + 20 + 50 + 30 + 20 + 10 + 6\right] = 149 \text{ cumecs}$$

$$\Delta t = 12 \text{ hours} = 12 \times 3600 \text{ seconds} = 43{,}200 \text{ s}$$

$$V_{total} = 149 \times 43{,}200 = 6{,}436{,}800 \text{ m}^3$$

Volume of Surface Runoff
Using the Direct Runoff values ordinates from the table:

$$\sum Q_{dr} = \left[\frac{0+0}{2} + 3 + 15 + 45 + 25 + 15 + 5 + 1\right] = 109 \text{ cumecs}$$

$$V_{surface} = 109 \times 43{,}200 = 4{,}708{,}800 \text{ m}^3$$

7. **What is a surge tank? Mention its functions. Also discuss with reasons the appropriate location for a surge tank. Present a sketch to clarify your answer. [5]**

Surge tank is a water storage device used as pressure neutralizer in hydropower water conveyance system to resists excess pressure rise and pressure drop conditions.

The primary **functions** of a surge tank include:

→ **Water Hammer Relief:** It absorbs sudden high-pressure rises, known as **water hammer**, caused by rapid valve closure or a sudden decrease in turbine load.

→ **Pressure Regulation:** It manages negative pressure and prevents vacuum formation when a valve is suddenly opened or turbine load increases.

→ **Flow Stabilization:** It helps regulate turbine speed by establishing flow equilibrium more quickly, thereby ensuring stable power output.

→ **Storage Reservoir:** It serves as a small temporary storage unit to meet brief fluctuations in water demand.

Appropriate Location and Reasons

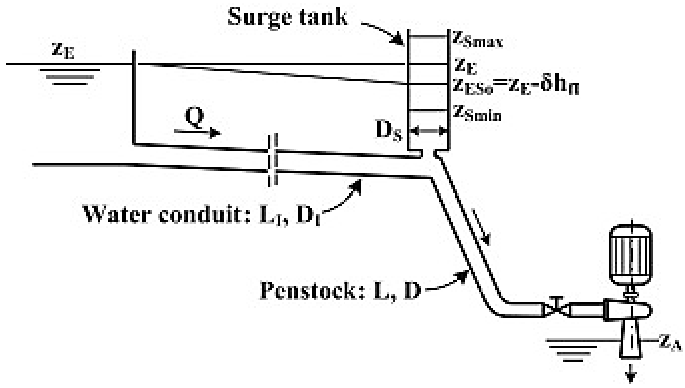
The ideal location for a surge tank is at the junction of the long low-pressure headrace tunnel and the short high-pressure penstock, as close to the powerhouse as possible.

Reasons for this location:

→ **Reducing Penstock Length:** Locating it near the powerhouse minimizes the length of the penstock, which must be designed to withstand high water hammer pressures.

→ **Isolating High Pressure:** Only the short section between the tank and turbine (the penstock) needs to resist high-pressure surges.

→ **Rapid Response:** Placing it as close to the powerhouse as possible ensures it can provide or absorb water almost instantly during sudden turbine load changes.



Section C

8. Define tack coat, seal coat and priming. Discuss the types of seal coat and process of their application. State in brief the classification of road maintenance activities in Nepal. [3+4+3]

- **Prime Coat:** An application of low-viscosity liquid bituminous material to an **absorptive surface** like a non-bituminous base course (e.g., WBM).
- **Tack Coat:** A very light application of liquid bitumen applied to an **existing bituminous or concrete surface**. Acts as an adhesive to ensure a strong bond between two asphalt layers.
- **Seal Coat:** A thin surface treatment applied as the **final layer** of the road. It serves to waterproof the surface, prevent the entry of air/moisture into the voids, and improve skid resistance.

Types of Seal Coat

Seal coats are classified based on the materials used and the texture provided:

a) **Sand Seal:** An application of bitumen followed by a thin layer of clean sand.

b) **Chip Seal (Single/Double Bituminous Surface Treatment - SBST/DBST):** A layer of bitumen followed by a layer of aggregate chips. In Nepal, **DBST** is very common for rural and secondary roads.

Process of Application

The general application process (specifically for a Chip Seal/DBST) involves:

→ **Surface Preparation:** The existing surface is cleaned of dust, mud, and loose debris using power brooms or air blowers.

→ **Heating and Spraying:** Bitumen is heated to the required temperature (75--85°C) for emulsions or higher for penetration grades) and sprayed uniformly using a pressure distributor.

→ **Aggregate Spreading:** Immediately after spraying, stone chips are spread evenly using a mechanical spreader.

→ **Rolling:** A pneumatic-tired roller is used to embed the chips into the binder. Rolling starts from the edges and moves toward the center.

→ **Curing and Finishing:** The road is closed to traffic for a few hours (or up to 24 hours) to allow the binder to set properly before excess chips are swept away.

Classification of Road Maintenance in Nepal

According to the **Department of Roads (DoR)** and the **Road Board Nepal (RBN)**, maintenance activities are classified into five main categories:

1) **Routine Maintenance:** Small-scale, continuous activities carried out year-round, such as cleaning drains, cutting grass, and clearing minor landslides.
2) **Recurrent Maintenance:** Periodic activities required a few times a year, such as pothole patching, crack sealing, and edge repair.
3) **Periodic Maintenance:** Major scheduled works carried out every few years (e.g., 5–7 years), such as applying a new **Seal Coat** or an asphalt overlay to restore the riding surface.
4) **Emergency Maintenance:** Immediate repairs required due to sudden failures, often caused by heavy monsoon rains, large landslides, or floods, to restore road connectivity.
5) **Specific Maintenance:** One-off repairs like building a new retaining wall or replacing a damaged culvert that does not fall under routine or periodically.

9. Explain the post card methods of O and D surveys. Discuss the advantage and disadvantage of the methods? how are the road intersection planned? Enumerated the various traffic controls needed at an intersection. [10]

The **Postcard Method** is an indirect technique used to collect Origin-Destination data. It is primarily used when traffic volumes are too high for roadside interviews, or when budget/personnel are limited.

- ✓ **Process:** Survey personnel stand at "cordon points" (entry/exit points of a study area). As vehicles slow down, drivers are handed a previously prepared reply postcard containing a brief questionnaire.
- ✓ **Survey Data:** Typically includes trip origin, destination, trip purpose (work, social, etc.), and the route taken.
- ✓ **Return:** The driver fills out the card at their convenience and mails it back.

Advantages vs. Disadvantages

Advantages	Disadvantages
No Traffic Disruption: Unlike roadside interviews, vehicles are only stopped for a few seconds to receive the card, minimizing congestion.	**Low Response Rate:** Typically, only 10–25% of cards are returned, which can lead to statistically insignificant results.
Lower Cost: Requires fewer personnel and less equipment on-site compared to household or roadside interviews.	**Sampling Bias:** Only "responsible" or interested citizens tend to reply, which may not accurately represent the entire driving population.
Convenience: Respondents can fill it out at home, leading to potentially more accurate address information.	**Delayed Data:** It takes several weeks to collect all the mailed responses before analysis can begin.

Planning of Road Intersections

Planning an intersection involves balancing **capacity** (moving vehicles) with **safety** (reducing accidents). The planning process generally follows these steps:

a) **Evaluation of Existing Conditions:** Collecting traffic volume counts (turning movements), identifying existing lane use, and checking for roadside hazards.
b) **Conflict Point Analysis:** Mapping where vehicle paths merge, diverge, or cross.

c) **Selection of Design Vehicle:** Determining the largest vehicle that will frequently use the junction to ensure turn radii are adequate.
d) **Geometric Layout:** Designing the angle of the intersection (ideally 90°) or close to it.
e) **Channelization:** Planning "islands" or medians to guide traffic into definite paths and provide refuge for pedestrians.

Traffic Controls at Intersections

Traffic controls are categorized into three levels of "strictness" based on the traffic volume and complexity of the intersection:

A. Passive Controls (Low Volume)

- **Right-of-Way Rules:** No signs; drivers follow basic rules (e.g., yield to the vehicle on the right).
- **Traffic Signs: STOP** signs (used when visibility is poor) or **YIELD/GIVE WAY** signs (used when visibility is good).
- **Road Markings:** Stop lines, yield triangles, and directional arrows.

B. Semi-Controls (Medium Volume)

- **Channelization:** Using traffic islands, curbs, and raised medians to separate conflicting movements and reduce the area of the conflict zone.
- **Traffic Rotaries (Roundabouts):** Converting all crossing conflicts into merging and diverging movements by forcing traffic around a central island.

C. Active Controls (High Volume)

- **Traffic Signals:** Using timed light phases (Red, Amber, Green) to alternate the right-of-way between competing movements.
- **Grade Separation:** The most extreme control—using flyovers, underpasses, or interchanges to eliminate physical intersection entirely.
- **Manual Control:** Traffic police officers directing flow (very common in high-traffic urban centers like Kathmandu).

10. Describe briefly about International Civil Aviation Organization (ICAO)'s requirement for aerodrome design and operations.[5]

The **International Civil Aviation Organization (ICAO)** sets the global blueprint for airport safety and efficiency through **Annex 14** to the Convention on International Civil Aviation.

These requirements are divided into **Standards** (mandatory compliance) and **Recommended Practices** (desirable for safety/efficiency).

Aerodrome design requirements:

→ Aerodromes must be designed to ensure **safety, regularity, and efficiency** of aircraft operations.
→ ICAO specifies standards for **runway length and width, runway strips, shoulders, taxiways, aprons, and obstacle limitation surfaces (OLS)**.
→ Requirements also cover **pavement strength (PCN), drainage, lighting systems, markings, signage, and visual aids**.
→ Separation distances between runways, taxiways, and obstacles are defined to minimize collision risks.

Aerodrome operation requirements:

→ Aerodromes must establish proper **operational procedures**, including runway inspections, wildlife hazard management, and control of obstacles.
→ **Aeronautical information**, such as runway conditions and declared distances, must be accurately published.

→ ICAO requires **emergency planning**, including rescue and fire-fighting services (RFFS) appropriate to the aerodrome category.
→ A **Safety Management System (SMS)** must be implemented to identify hazards and manage operational risks.
→ Regular maintenance of movement areas and visual aids is mandatory to ensure continued compliance.

Section D

11. Why is water treatment necessary in water supply system? Explain the different methods of removal of iron and manganese in ground water that is common in Kathmandu Valley and in Terai region. [2+4+4]

Water treatment is essential in supply systems for the following reasons:
→ **Elimination of Pathogenic Microorganisms:** It removes or kills harmful bacteria, viruses, and protozoa (such as those causing Cholera and Typhoid).
→ **Removal of Toxic Minerals (like Iron, Manganese**, and **Arsenic** above the certain limits)
→ **Improvement of Aesthetic Quality:** Treatment processes eliminate cloudiness (turbidity), foul odors (like the "rotten egg" smell of hydrogen sulfide).
→ **Compliance and Safety Standards:** It ensures that the water delivered to the public consistently meets health safety regulations as per National Drinking Water Quality Standards (NDWQS).

In the Kathmandu Valley and the Terai region of Nepal, groundwater is a primary source of water but frequently contains high levels of dissolved iron (Fe^{2+}) and manganese (Mn^{2+}). These minerals cause issues like reddish-brown or black staining and metallic tastes.

The presence of Iron and manganese in water up to **0.3 mg/lit for Iron and 0.2 mg/lit for Manganese** is tolerable as per National Drinking Water Quality Standards (NDWQS), and no treatment is needed below these limits.

Most common treatment methods used specifically in these regions of Nepal based on standard practices and environmental engineering studies.

1. Aeration followed by Filtration

This is the **most common method** used in municipal water treatment plants (WTPs) throughout the Kathmandu Valley (e.g., Mahadevkhola, Bhaktapur) and the Terai.

Aeration: Water is exposed to air (using cascade aerators, tray aerators, or air compressors). This introduces dissolved oxygen which oxidizes soluble ferrous iron (Fe^{2+}) into insoluble ferric hydroxide $Fe(OH)_3$ precipitates.

The oxidized iron and manganese precipitate (form solid particles) and can then be removed by filtration through sand or other media.

2. Oxidation and Filtration:

Like aeration, this method uses chemical oxidants like chlorine or potassium permanganate to convert dissolved iron and manganese into removable precipitates.

3. Ion Exchange:

This process uses specialized resins that exchange iron and manganese ions for other harmless ions like sodium.

4. Greensand Filtration:

This method utilizes a specially coated sand media that acts as a catalyst for iron and manganese oxidation and filtration in a single step.

12. **What is lagooning? Write about its advantages and disadvantages. Also, compare lagooning with dumping. [5+5]**

Lagooning (also known as a **Waste Stabilization Pond** or **Oxidation Pond**) is a low-cost, natural wastewater treatment process that uses large, shallow, man-made basins to treat sewage or industrial waste.

Advantages and Disadvantages of Lagooning

Advantages	**Disadvantages**
Low Cost: Cheap to construct (mostly earthwork) and requires very little energy to operate.	**Large Land Requirement:** Requires significant surface area, making it unsuitable for urban or crowded regions.
Simple Operation: Does not require highly skilled technical staff or complex mechanical parts.	**Weather Dependency:** Efficiency drops in cold climates or during seasons with low sunlight/high rainfall.
High Pathogen Removal: Effectively reduces bacteria and viruses due to long retention times and UV exposure.	**Odor Issues:** If overloaded or poorly maintained, lagoons can release foul "rotten egg" smells (hydrogen sulfide).
	Groundwater Risk: If not properly lined with clay or plastic, untreated waste can seep into the water table.

Comparison: Lagooning vs. Dumping

Feature	**Lagooning (Treatment)**	**Dumping (Disposal)**
Objective	To **treat and purify** water before returning it to the environment.	To **discard** waste without any intention of treatment.
Design	Scientifically engineered with specific depths, liners, and controlled flow.	Often unstructured; waste is simply piled or poured into a hole or water body.
Environmental Impact	Protects the environment by reducing pollutants and pathogens.	Highly polluting; leads to water contamination, soil degradation, and disease.
Regulation	Subject to strict environmental standards and monitoring.	Usually illegal or unregulated "open dumping."
Process	Uses natural biological stabilization (aerobic/anaerobic processes).	Relies on raw decomposition, which is often messy, smelly, and hazardous.

THE END

MODEL SET 12

लोक सेवा आयोग

नेपाल इन्जिनियरिङ सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह,

राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

समय: ३ घण्टा पूर्णाङ्क: १००

पत्र: द्वितीय विषय: Technical Subject

तलका प्रश्नहरुको उत्तर Section अनुसार छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ।

Section A

1. **Write down the design steps for doubly reinforced beam section. [5]**

In a doubly reinforced beam, the depth (d) is usually restricted or known, and the width (b) is selected using the ratio of D/B (1:2 to 1:3)

Step 1: Calculate the maximum moment a balanced singly reinforced section can carry using the following formulas based on the grade of steel:

For Fe250: $M_{ulim} = 0.148 f_{ck} b d^2$

For Fe415: $M_{ulim} = 0.138 f_{ck} b d^2$

For Fe500: $M_{ulim} = 0.133 f_{ck} b d^2$

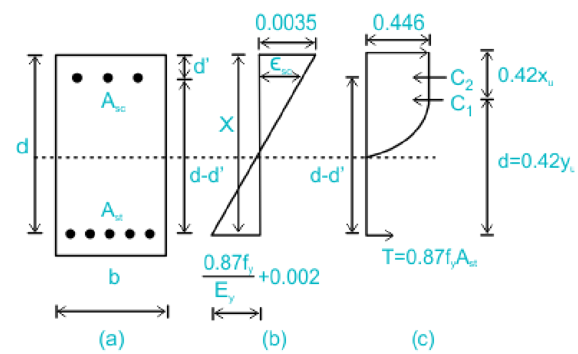


Fig 1: Doubly reinforced beam

Step 2: If $\mathbf{M_u > M_{ulim}}$, the section must be designed as a **doubly reinforced section**.

Step 3: Find the area of tensile steel required to resist the limiting moment M_{ulim} as a balanced singly reinforced section equating,

$$0.36 f_{ck} b\, x_{ulim} = 0.87 f_y A_{st1}$$

Step 4: Find the surplus moment M_{u2} that the compression steel and additional tensile steel must carry:

$$M_{u2} = M_u - M_{ulim}$$

$$M_{u2} = 0.87 f_y A_{st2} (d - d')$$

Where d' is the effective cover to the compression steel.

Step 5: Calculate Compression Steel (A_{sc})

Determine the strain in compression steel (ϵ_{sc}) using the linear stress-strain diagram:

$$\epsilon_{sc} = 0.0035 \frac{x_{ulim} - d'}{x_{ulim}}$$

Find the corresponding stress (f_{sc}) from the strain value. Then, calculate the area of compression steel by equating forces:

$$f_{sc} A_{sc} = 0.87 f_y A_{st2}$$

Step 6: Design of Shear Reinforcement

- Determine Nominal Shear Stress (τ_v): $\tau_v = \frac{V_u}{bd}$
- Determine Design Shear Strength (τ_c): Find the design shear strength of concrete from the code.
- Design Criteria Based on Shear Stress Values:
  - If $\tau_v < \tau_c$: Provide only nominal shear reinforcement.
  - If $\tau_v > \tau_c$: Design, the specific shear reinforcement required.
  - If $\tau_v > \boldsymbol{\tau_{cmax}}$: The section is inadequate and must be revised.

2. What do you mean by economical span length of a bridge? Explain. What types of loads are required to be considered while designing a road bridge? [3+7]

The span of the bridge for which the total cost of the bridge is minimum is called economic span.
Let, the ratio of cost of one span 'S' to cost of one pier 'P' be R (i.e. R=S/P).
Then,
The overall cost of the bridge considered be C = nS + (n - 1) * P
Putting, S=RP
C = n(RP) + (n - 1) * P
where, n= Number of spans and (n - 1) = Number of piers
For C to be minimum, d/dn (C) = 0
Hence, S = P
i.e. economical span is one for which the cost of one span is equal to one pier.
In other words, Economical span is defined as that span for which the total cost of the substructure is equal to the total cost of Super Structure.

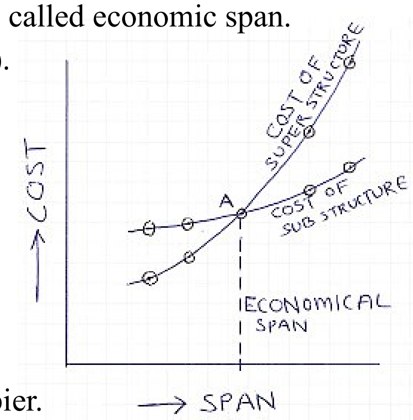


Fig 2: Economic span of bridge

Second Part:

When designing a road bridge, several types of loads must be considered, these loads are generally categorized as follows:

1. Permanent Loads

These are loads that act on the bridge throughout its entire lifespan.

→ **Dead Load (D.L):** The self-weight of the structural elements, such as the deck slab, beams, girders, and piers.

→ **Superimposed Dead Load:** The weight of non-structural components added after the main structure is complete, including the wearing surface (asphalt), railings, sidewalks, and utility pipes.

2. Temporary (Live) Loads

These are forces that move across the bridge.

→ **Vehicular Live Load:** The weight of cars, trucks, and other vehicles. Standardized "design vehicles" such as **IRC Class AA, 70R, Class A,** and **Class B** loadings.

→ **Impact Load:** Dynamic effect of moving vehicles, such as bouncing or vibrations caused by road surface irregularities.

→ **Pedestrian Load:** The weight of people walking on footpaths or the main deck.

3. Lateral and Longitudinal Forces

Forces acting horizontally or along the length of the bridge.

→ **Longitudinal Forces:** Vehicles suddenly braking/accelerating

→ **Centrifugal Forces:** When vehicles move along a curved bridge path

→ **Wind Load:** The pressure exerted by wind on the bridge's superstructure and substructure.

→ **Earth Pressure:** The pressure exerted by soil against the bridge abutments and retaining walls.

4. Environmental and Environmental Loads

a) **Water Current and Buoyancy:** Horizontal pressure from flowing river water and the upward force (buoyancy) on submerged bridge piers.

b) **Seismic (Earthquake) Loads:** Due to ground movement during an earthquake.

c) **Thermal Effects:** Stresses caused by the expansion or contraction of bridge materials due to temperature changes.

3. **The time for a clay layer to achieve 99% consolidation is 10 years. What time would be required to achieve 99% consolidation if the layer were twice a thick, five times more permeable and three times more compressible? [5]**

$t_1 = 10 \text{ years}, h_2 = 2 \times h_1, k_2 = 5 \times k_1, m_{v2} = 3 \times m_{v1}$

$T_v = \dfrac{C_v \times t}{h^2} \rightarrow t = \dfrac{T_v \times h^2}{C_v}$ Also, $C_v = \dfrac{k}{\gamma_m \times m_v}$

$\therefore \dfrac{t_2}{t_1} = \left(\dfrac{h_2}{h_1}\right)^2 \times \dfrac{k_1}{k_2} \times \dfrac{m_{v2}}{m_{v1}} = \dfrac{4}{1} \times \dfrac{1}{5} \times \dfrac{3}{1} = \dfrac{12}{5} \rightarrow t_2 = \dfrac{12}{5} \times 10 = 24 \text{ years}$

4. **Define active and passive earth pressure in soil. Derive an expression for active and passive earth pressure by Rankine's method. [10]**

Active Earth Pressure: Active earth pressure is the minimum lateral force exerted by soil when a retaining structure moves away from the backfill.

Condition: Wall moves away from soil.

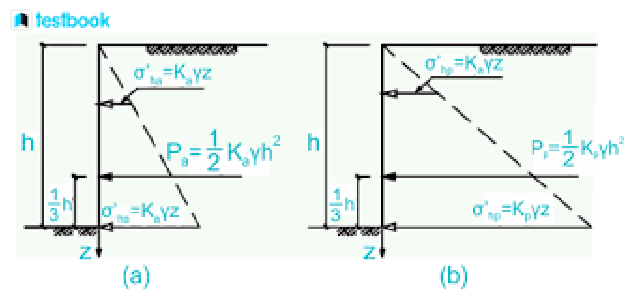
Passive Earth Pressure: Passive earth pressure is the maximum resistance force when a structure moves into the backfill, compressing the soil to failure.

Condition: Wall moves toward the soil.

Derivation by Rankine's Method

Rankine's theory assumptions:

a) The soil is homogeneous, isotropic and semi-infinite.
b) The backfill surface is horizontal.
c) Backface of wall is vertical and smooth.
d) Mohr-Coulomb failure criterion is valid.



General Relationship

For a soil with internal friction angle ϕ, the relationship between the major principal stress (σ_1) and minor principal stress (σ_3) at failure is:

$$\sigma_1 = \sigma_3 \tan^2\left(45^\circ + \frac{\phi}{2}\right) + 2c\tan\left(45^\circ + \frac{\phi}{2}\right)$$

For cohesionless soil (sand), c = 0, the equation simplifies to:

$$\sigma_1 = \sigma_3\left(\frac{1 + \sin\phi}{1 - \sin\phi}\right)$$

A. Active Earth Pressure Expression

In the active state, the vertical stress $\sigma_v = \gamma z$ is the major stress (σ_1), and the active lateral pressure p_a is the minor stress (σ_3).

Substitute $\sigma_1 = \gamma z$ and $\sigma_3 = p_a$ into the failure equation:

$$\gamma z = p_a \tan^2\left(45^\circ + \frac{\phi}{2}\right)$$

$$p_a = \gamma z \frac{1}{\tan^2\left(45^\circ + \frac{\phi}{2}\right)} = \gamma z \tan^2\left(45^\circ - \frac{\phi}{2}\right)$$

Also, **Coefficient of Active Earth Pressure (K_a):**

$$K_a = \tan^2\left(45^\circ - \frac{\phi}{2}\right) = \frac{1 - \sin\phi}{1 + \sin\phi}$$

Final Expression: $p_a = K_a \gamma z$

B. Passive Earth Pressure Expression

In the passive state, the horizontal pressure p_p is pushed until it becomes the major stress (σ_1), while the vertical stress $\sigma_v = \gamma z$ remains the minor stress (σ_3).

Substitute $\sigma_1 = p_p$ and $\sigma_3 = \gamma z$ into the failure equation:

$$p_p = \gamma z \tan^2\left(45^\circ + \frac{\phi}{2}\right)$$

Also, **Coefficient of Passive Earth Pressure (K_p):**

$$K_p = \tan^2\left(45^\circ + \frac{\phi}{2}\right) = \frac{1 + \sin\phi}{1 - \sin\phi}$$

Final Expression: $p_p = K_p \gamma z$

Section B

5. **Describe the derivation of a unit hydrograph from a direct runoff hydrograph. Then show how you would transform it to a different duration. [10]**

A **Unit Hydrograph (UH)** represents the direct runoff resulting from a unit depth of effective rainfall occurring uniformly over a watershed for a specific duration.

Derivation of a Unit Hydrograph (UH) from a Direct Runoff Hydrograph (DRH) involves following steps:

a) **Separate Baseflow and find DRH Ordinates:** Subtract the base flow from the observed streamflow hydrograph to obtain the DRH ordinates.
b) **Calculate Total Runoff Volume:** Determine the volume of water using the DRH ordinates.
c) **Find effective Runoff Depth (d):** Divide the total runoff volume by the watershed area to find the depth of effective rainfall.
d) **Calculate UH Ordinates:** Divide each ordinate of the DRH by the calculated depth (d) to obtain the UH ordinates.

Second Part:

To convert a unit hydrograph of duration D to a different duration D', two primary methods are used based on the relationship between the durations:

1. Superposition Method (for integer multiples)

Use this when the new duration D' is an integer multiple of the original duration $D'=n\times D$.

Step 1: Lag the original D-hour UH n times, with each lag being D hours.

Step 2: Sum the ordinates of these n lagged hydrographs.

Step 3: Divide the resulting summed ordinates by n to get the new D'-hour UH.

2. S-Curve Method (for any duration)

Use this when the ratio D'/D is not an integer.

Step 1: Construct the S-Curve: Lag several D-hour UHs by D hours and sum them continuously until the flow reaches a constant equilibrium value.

Step 2: Shift the S-Curve: Offset a second identical S-curve by the desired new duration D'.

Step 3: Find the Difference: Subtract the ordinates of the lagged S-curve from the original S-curve.

Step 4: Scale to Unit Depth: Multiply the resulting difference by (D/D') to normalize it to a 1-unit depth for the new duration.

6. **What factors should be considered while formulating an irrigation scheme? What are the modes of irrigation? [5]**

Key factors that need to be considered while formulating an effective irrigation scheme are:

a) Water Resources Factors: Source & Availability, Water Quality, Water Rights and Legislation.
b) Agronomic and Land Factors: Soil characteristics, Crop Water Requirements, Crop Pattern & Rotation, Topography of CA
c) Climatic Factors: Rainfall and Evapotranspiration
d) Technical and Infrastructural Factors: Irrigation method selection, System design and efficiency,
e) Economic & Financial Factors: Capital costs, O&M Costs, Financial viability
f) Social, Institutional & Managerial Factors: Farmer Participation & Acceptance

<u>**Modes of Irrigation: -**</u>

1. Surface Irrigation:

This is the traditional and most common method. Water is allowed to flow over the soil surface by gravity to wet the field.

Examples:

Free Flooding/Check Flooding: Water is spread over the entire field (e.g., for paddy).

Furrow: Water flows through small channels between crop rows (e.g., for maize, sugarcane).

Basin: Water is confined to a level area around a tree or a bed of plants.

Advantage: Low cost, simple.

Disadvantage: Water wastage, needs leveled land.

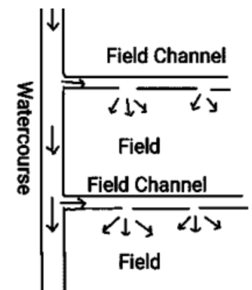


Fig: Free flooding

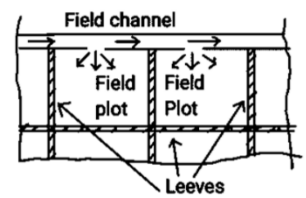


Fig: Check flooding

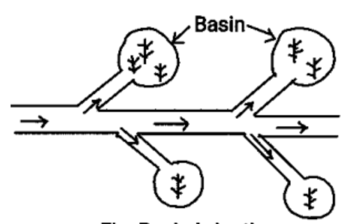


Fig: Basin Irrigation

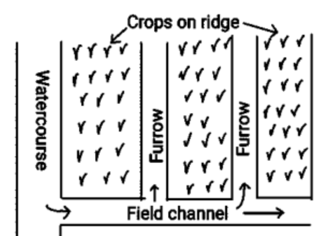


Fig: Furrow irrigation

2. Sprinkler Irrigation:

Here, water is sprayed into the air like natural rainfall using a system of pipes and pumps.

How it works: Water under pressure passes through nozzles, creating a spray over the crop.

Advantage: Good for uneven land, saves water compared to flooding.

Disadvantage: High initial cost, affected by strong wind.

3. Drip/Trickle Irrigation:

Water is applied drop by drop directly to the root zone of the plant through a network of pipes and emitters.

How it works: A pressurized system delivers water slowly and precisely at the plant base.

Advantage: Maximum water saving, reduces weeds, suitable for all terrains.

Disadvantage: High cost, emitters can clog, requires careful management.

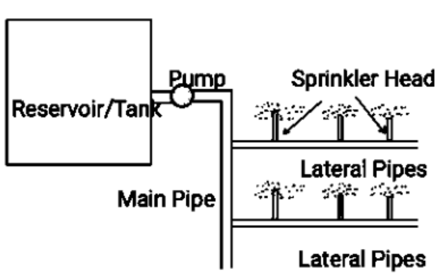


Fig: Sprinkler Irrigation

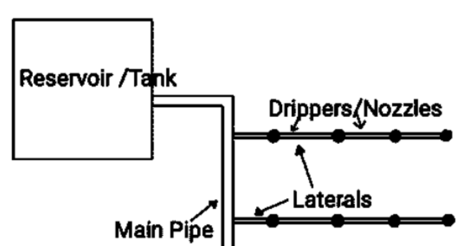


Fig: Drip Irrigation

7. **What do you mean by concrete gravity dams? With a neat sketch, show the different forces acting on it. How do you check the stability of dam against sliding and overturning? [2+3+5]**

A **concrete gravity dam** is a massive, solid structure constructed from Plain cement concrete or RCC. Its name "gravity" comes from its primary mode of stability: it **resists the horizontal force of the retained water entirely by its own weight**.

<u>Forces acting on Gravity Dam:</u>

Primary Forces:

Water pressure

Uplift pressure or seepage loads

Self-weight of the dam

Secondary Forces:

Silt pressure

Wave pressure

Wind Pressure

Ice pressure

Exceptional Forces - Earthquake forces

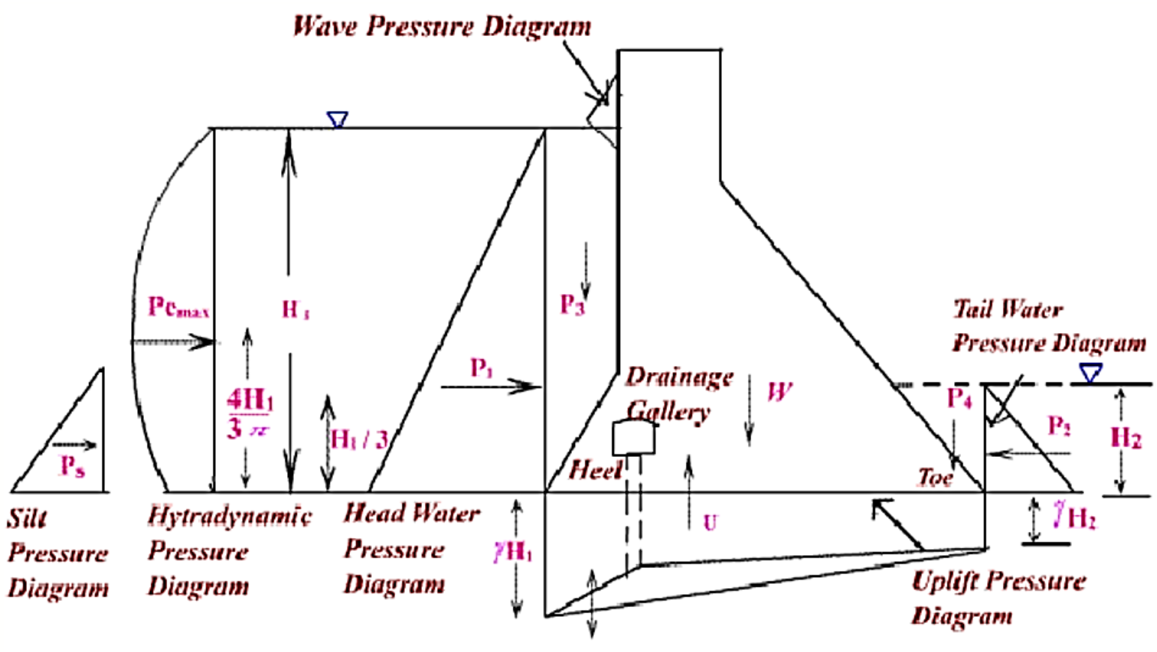


Fig 3: Forces acting on Gravity Dam

Checking Stability

1. Stability Against Overturning

A dam can tip over its "toe" if the rotating moment caused by the water pressure is greater than the moment due to dam's weight.

$$FOS_o = \frac{\sum M_R}{\sum M_O}$$

$\sum \mathbf{M_R}$ **(Restoring Moments):** Moments that resist tipping (primarily from the dam's weight).

$\sum \mathbf{M_O}$ **(Overturning Moments):** Moments that try to tip the dam (primarily from water and uplift pressure).

Requirement: Usually, FOS_o must be **greater than 1.5**.

2. Stability Against Sliding

Sliding occurs when the horizontal forces (water pressure) exceed the frictional resistance between the dam and its foundation.

$$FOS_s = \frac{\mu \cdot \sum V}{\sum H}$$

μ: The coefficient of friction between the concrete and the foundation.
∑ V: The net vertical force (Weight - Uplift).
∑ H: The total horizontal force (Water + Silt + Wave pressure).
Requirement: Generally, FOS_s should be **greater than 1.0**

Section C

8. What is belt and road initiative program? what may its advantages and disadvantages for Nepal? [10]

The **Belt and Road Initiative (BRI)**, originally known as "One Belt, One Road," is a massive global infrastructure and economic development strategy launched by China in 2013. It aims to revitalize the ancient Silk Road trade routes through two main components:

- **The Silk Road Economic Belt:** Land-based corridors connecting China with Central Asia, Europe, and South Asia.
- **The 21st Century Maritime Silk Road:** Sea-based routes connecting China to Southeast Asia, Africa, and Europe via the Indian Ocean.

Nepal formally joined the BRI on **May 12, 2017**, by signing a Memorandum of Understanding (MoU). For a landlocked country like Nepal, the BRI is seen as a "paradigm shift" from being landlocked to becoming a **land-linked** nation.

Advantages for Nepal

a) **Enhanced Connectivity:** The most significant project, the **Trans-Himalayan Multi-Dimensional Connectivity Network**, includes roads, railways (Kerung-Kathmandu), and transmission lines. This will provide Nepal with an alternative transit route, reducing its total dependency on India.
b) **Infrastructure Modernization:** BRI projects focus on large-scale infrastructure like the Tokha-Chhahare Tunnel and cross-border power transmission lines (220 kV, Jilong-Rasuwagadhi-Chilime), which are essential for long-term economic growth.
c) **Trade and Market Access:** Improved transport links can reduce the cost and time of trade. Nepal could find new markets for its agricultural and herbal products in China’s Tibet Autonomous Region and beyond.
d) **Boost to Tourism:** Better Road and rail links are expected to increase the number of Chinese tourists visiting Nepal, providing a vital source of foreign exchange.
e) **Soft Infrastructure:** Beyond physical roads, the BRI includes the **Digital Silk Road** and educational scholarships, helping develop Nepal's technical workforce through institutions like the Madan Bhandari University of Science and Technology.

Disadvantages and Challenges for Nepal

a) **Debt Trap Concerns:** Critics point to countries like Sri Lanka (Hambantota Port) as cautionary tales. Nepal's economy is small, and there is a fear that high-interest commercial loans for mega-projects could lead to an unsustainable debt-to-GDP ratio.
b) **Geopolitical Sensitivity:** Nepal must balance its relations with both China and India. India has stayed away from the BRI and views China's growing influence in its neighborhood with suspicion, which can lead to diplomatic friction.
c) **Project Implementation Delays:** Despite signing the agreement in 2017, progress has been slow. Issues regarding funding (Nepal prefers grants or soft loans, while China often offers commercial loans) have stalled several projects.
d) **Environmental Risks:** Constructing railways and highways through the fragile Himalayan ecosystem poses significant environmental threats, including landslides, biodiversity loss, and impact on water sources.
e) **Lack of Transparency:** Many critics argue that the terms of BRI agreements are often confidential, leading to public concern over sovereignty and the actual "win-win" nature of these deals.

9. **What do you understand by low-cost pavement? Explain application of Otta seal. [5]**

Low-cost pavement refers to road that prioritizes the use of local materials and simplified construction methods to reduce initial capital investment. Generally, designed for low to medium traffic volumes.

An **Otta Seal** is a specific type of low-cost pavement consisting of a mixture of bitumen and graded **aggregates** (such as natural gravel from a riverbed or a quarry).

How it Works:

→ The bitumen is sprayed on the base, and the graded aggregate is spread over it.
→ Traffic is then allowed to drive over it immediately.
→ The weight of the vehicles pushes the various sizes of stones into the bitumen, creating a dense, matrix-like structure similar to a thin layer of asphalt.

Applications of Otta Seal

In Nepal, the Department of Local Infrastructure (DoLI) often favors Otta Seals for mountainous terrain because the process is less dependent on heavy machinery.
(Source: International Focus Group on Rural Road Engineering by DoR)
Otta seals should be considered when;
→ a road is to be constructed in a remote area where only natural gravels occur;
→ workmanship may be of indifferent quality;
→ flexibility and durability are required to tolerate low quality, low bearing capacity pavements; or
→ there is a low maintenance capability.

10. **What are the factors to consider in site selection of Airport? Discuss different types of maintenance operations for flexible pavements. [5+5]**

Key factors that must be considered during site selection:
→ **Regional plan:** Alignment with the broader development goals of the area.
→ **Airport use:** Determining whether the airport is for commercial, military, or private use.
→ **Proximity to other airports:** Ensuring there is no airspace conflict or redundant service.
→ **Ground accessibility:** Ease of transport for passengers and cargo to and from the city.

→ **Topography:** The physical features and "lay of the land."
→ **Obstructions:** Identifying natural or man-made hazards like mountains or tall buildings.
→ **Visibility:** Assessing prevalence of fog, smoke, or haze that could affect flight operations.
→ **Wind:** Direction and intensity, which are critical for runway orientation.
→ **Noise nuisance:** Impact on surrounding residential or sensitive areas.
→ **Grading, drainage, and soil characteristics:** Technical requirements for stable construction and water management.
→ **Future development:** Potential for expanding the airport as demand grows.
→ **Availability of utilities from town:** Access to water, electricity, and telecommunications.
→ **Economic considerations:** Cost-benefit analysis, including land acquisition and construction expenses.

Second Part:
Here are the five primary types of maintenance operations:
1. Routine Maintenance
This is a continuous activity carried out throughout the year.
Key Activities: Cleaning drains and culverts, clearing minor landslides, cutting grass on shoulders, and cleaning road signs.
2. Recurrent Maintenance
These operations are performed a few times a year, usually before and after the monsoon season.
Key Activities: Patching small potholes, sealing minor cracks (crack sealing), and repairing edge breaks.
3. Periodic Maintenance
This involves larger-scale works carried out at intervals of several years (typically every 5–8 years for paved roads in Nepal).
Key Activities: Applying a new surface dressing, thin overlays (like asphalt concrete), or resealing the entire road surface.
4. Specific (Special) Maintenance
Specific maintenance refers to one-off repairs or improvements that are not covered by routine or periodic schedules. These are often location-specific.
Key Activities: Repairing a collapsed retaining wall, significant gully repairs, or fixing a specific section of the road that has suffered localized subgrade failure.
5. Emergency Maintenance
This is immediate work required to restore traffic flow after sudden, unforeseen events.
Key Activities: Removing large landslides that block the road, repairing washed-out sections after heavy flooding, and clearing debris after an earthquake.

Section D

11. Classify the sedimentation tasks. How can we increase the setting efficiency of particles? [5]
Types of Sedimentation Tank
Sedimentation tanks are classified by method of operation into:
1. Quiescent or Fill and Draw Type
→ Water is filled, allowed to settle for a long detention time (e.g., 24 hours), then cleared water is drawn out and sediment is cleaned manually.
2. Continuous Flow Type

→ Raw water flows continuously through inlet, slows down in the tank to allow settling, and cleared water exits continuously from outlet.
→ Works on the principle of reducing flow velocity to let particles settle.
→ It can be rectangular, circular, or square in shape.
Subtypes: **Horizontal flow type and Vertical flow type.**

Increasing the efficiency of particle setting in sedimentation tanks involves:
→ Optimize Tank Geometry:
L/B Ratio: Rectangular tanks should ideally have a length 3 to 5 times their width for uniform flow.
Increased Surface Area: Efficiency is primarily a function of surface area rather than just tank depth.
→ Control Horizontal flow velocity and maintain Laminar Flow.
→ Coagulation and Flocculation: Adding chemicals like aluminum sulfate (alum) or ferric chloride neutralizes the negative charge of fine particles, allowing them to clump into larger, denser "flocs" that settle faster.
→ Install Baffles: Baffles at the inlet dissipate incoming flow energy and distribute water uniformly across the tank.
→ Efficient Sludge Removal

12. What is self-cleaning velocity and no-scouring velocity in a sewer? Why are sewers not designed to flow full? Design a sewer to serve a population of 36000, the daily per capita water supply allowance being 135 liters, of which 10% find its way into the sewer. The slope available for the sewer to be laid is 1 in 625 and the sewer should be designed to carry 4 times the dry weather flow when running full. What would be the velocity of flow in the sewer when running full? Assume n = 0.012 in Manning's formula? [10]

Self-Cleaning Velocity: This is the minimum water flow velocity required to transport solids through the sewer pipe and prevent them from settling and accumulating.
Non-Scouring Velocity: This is the maximum water flow velocity that can safely travel through the sewer pipe without causing erosion or damage to the pipe lining.

Sewer systems are not designed to operate at full capacity for several reasons:
To Ventilate Sewer Gases: A space at the top allows harmful and explosive gases like methane and hydrogen sulfide to escape through designated vents, preventing dangerous build-up.
Accommodating Peak Flows: Sewer usage fluctuates throughout the day, with peak periods during showers, laundry cycles, etc. Designing for full flow would be wasteful for most of the day.
Preventing Surcharges and backflow: Full pipes can lead to backflow into homes and streets (sewer surcharges).
Optimal Hydraulic Efficiency: According to hydraulic principles, a circular pipe reaches its maximum discharge when it is approximately 93% full, and its maximum velocity at 81% full.

Numerical Portion:
1. Dry Weather Flow (DWF)
Population = 36,000
Water supply = 135 L/capita/day
Sewage reaching sewer = 10%

Manning's$(n = 0.012)$
Slope $(S = 1/625 = 0.0016)$

Sewage per capita $= 0.10 \times 135 = 13.5\ \text{L/day}$
Total sewage $= 36{,}000 \times 13.5 = 486{,}000\ \text{L/day} = 486\ \text{m}^3/\text{day}$
$\text{DWF} = \dfrac{486}{86{,}400} = 0.00563\ \text{m}^3/\text{s}$

2. Design Discharge
The sewer is designed to carry **4 times the DWF when running full**:
$Q = 4 \times 0.00563 = 0.0225\ \text{m}^3/\text{s}$

3. Use Manning's Formula (Running Full)

For a circular sewer running full:
Area $\left(A = \dfrac{\pi D^2}{4}\right)$
Hydraulicradius $\left(R = \dfrac{D}{4}\right)$
$Q = \dfrac{1}{n} A R^{2/3} S^{1/2}$
Substituting and solving gives:

$$D \approx 0.24\ \text{m}$$

4. Velocity When Sewer Runs Full

$$A = \frac{\pi}{4}(0.24)^2 = 0.045\ \text{m}^2$$
$$V = \frac{Q}{A} = \frac{0.0225}{0.045} \approx 0.50\ \text{m/s}$$

13. What do you understand by greenhouse effect? What are its causes? How it can be mitigated ?[5]
Greenhouse Effect: It's a natural process where certain gases in Earth's atmosphere (like carbon dioxide) act like a blanket, they let sunlight in but trap some of the heat that would otherwise escape back into space.

Main Causes: (Human Activities)
We are adding too many heat-trapping gases by:
→ Burning coal, oil, and gas for electricity, vehicles, and factories (releases CO_2).
→ Cutting down forests (trees that absorb CO_2 are removed).
→ Large-scale agriculture (livestock produce methane, fertilizers release nitrous oxide).
→ Landfills and industrial processes releasing other potent gases.

Mitigation: (Solutions)
We can reduce the impact by:
→ Switching to clean energy like solar and wind power.
→ Protecting and planting forests to absorb CO_2.

→ Using energy-efficient technologies and electric vehicles.

→ Improving waste management (e.g., capturing landfill methane, recycling).

→ Following strong climate policies and international agreements (like the Paris Agreement).

THE END

MODEL SET 13

लोक सेवा आयोग

नेपाल इन्जिनियरिङ सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह,

राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

समय: ३ घण्टा **पूर्णाङ्क: १००**

पत्र: द्वितीय **विषय: Technical Subject**

तलका प्रश्नहरुको उत्तर Section अनुसार छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ।

Section A

1. **Design a two-way slab for a room with a clear dimension of 5mx4m. The superimposed load 200 kg/m² using M15 mix and Fe415. Assume corners of the slabs are not held down. [10]**

1. Selection of Slab Depth

For a two-way slab simply supported, we use a span-to-effective depth ratio,

We have, l/d ratio $\leq \alpha\beta\gamma\delta\lambda$

Now, $\alpha =$ 20 and assuming 0.3% (range 0.2-0.3%) steel, $\beta = 1.43$

$$d = \frac{L_x}{20 \times 1.43} = 139.86 \text{ mm}$$

Adopt:

Overall depth $= 180$ mm

Effective cover ≈ 30 mm

$$d = 180 - 30 = 150 \text{ mm}$$

2. Loads on Slab

(a) Self weight $= 0.18 \times 25 = 4.5 \text{ kN/m}^2$

(b) Superimposed load $= 2.0 \text{ kN/m}^2$

(c) Total service load, $w = 4.5 + 2.0 = 6.5 \text{ kN/m}^2$

(d) Factored load, $w_u = 1.5 \times 6.5 = 9.75 \text{ kN/m}^2$

3. Bending Moment Calculation

From **IS 456 Table 26**, for

$\frac{L_y}{L_x} = 1.25$ and **corners not held down**:

$$\alpha_x = 0.086 \text{ and } \alpha_y = 0.056$$

(a) Moment in short span (x-direction)

$$M_x = \alpha_x w_u L_x^2$$
$$M_x = 0.086 \times 9.75 \times 4^2$$
$$M_x = 13.4 \text{ kNm/m}$$

(b) Moment in long span (y-direction)

$$M_y = \alpha_y w_u L_x^2$$
$$M_y = 0.056 \times 9.75 \times 4^2$$
$$M_y = 8.7 \text{ kNm/m}$$

4. Checking for Depth

Limiting moment capacity for Fe415:

$$M_{u,lim} = 0.138 f_{ck} b d^2$$
$$= 0.138 \times 15 \times 1000 \times 150^2$$
$$= 46.6 \text{ kNm/m}$$

Both M_x and M_y are **much less in comparison to** $M_{u,lim}$, so depth is safe.

5. Steel Reinforcement Design

(a) Short span reinforcement (main steel)

Using:

$$M_u = 0.87 f_y A_{st} d \left(1 - \frac{A_{st} f_y}{f_{ck} bd}\right)$$

$$\text{or, } 13.4 \times 10^6 = 0.87 \times 415 \times A_{stx} \times 150 \left(1 - \frac{A_{stx} \times 415}{25 \times 1000 \times 150}\right)$$

$$\therefore A_{stx} \approx 255 \text{ mm}^2/\text{m}$$

Provide: **10 mm Ø bars @ 150 mm c/c**

$$A_{prov} = 523 \text{ mm}^2/\text{m}$$

(b) Long span reinforcement (secondary steel)

Required:

$$M_u = 0.87 f_y A_{st} d \left(1 - \frac{A_{st} f_y}{f_{ck} bd}\right)$$

$$\text{or, } 8.7 \times 10^6 = 0.87 \times 415 \times A_{sty} \times 150 \left(1 - \frac{A_{sty} \times 415}{25 \times 1000 \times 150}\right)$$

$$\therefore A_{sty} \approx 164 \text{ mm}^2/\text{m}$$

Provide: **8 mm Ø bars @ 150 mm c/c**

$$A_{prov} = 335 \text{ mm}^2/\text{m}$$

6. Spacing Check (IS 456)

Maximum spacing allowed: $\leq \min(3d, 300) = 300 \text{ mm}$

Provided spacing = **150 mm** (Safe)

7. Torsional Reinforcement

Since corners are not held down. So, no torsional reinforcement required

2. **What are the requirements of earthquake-resistant building construction? Describe. [5]**

The requirements for earthquake-resistant building construction are categorized into several key structural and design factors such as:

1. Site Selection

The location and the ground on which a building stands are critical:

→ **Stable Ground:** Foundations must be built on stable, firm ground.

→ **Topography:** The site should be free from areas prone to landslides.

2. Building Shape and Geometry

The geometry of a building significantly affects its stability during seismic activity.

→ **Regular Shapes:** Buildings with regular shapes (e.g., **rectangular** or **square**) are considered good for earthquake resistance.

→ **Re-entrant Corners:** Shapes with "re-entrant corners" (like **L-shaped** or **U-shaped** buildings) are considered unsafe because they create stress concentrations at the corners.

→ **Symmetry:** Uniform geometry helps in distributing seismic forces evenly.

→ **Uniformity:** Columns of different heights should be avoided to prevent irregular load distribution.

3. Foundation Depth

Minimum depth requirements based on the type of soil:

→ **Clayey Soil:** Minimum foundation depth should be **900 mm**.

→ **Hard Rock:** Minimum foundation depth should be **450 mm**.

4. Masonry Work

Proper masonry techniques ensure the walls act as a cohesive unit:

→ Stones should be placed with the breadth side exposed.

→ Through Stones should be used at regular intervals (after a certain height) to bind the inner and outer layers of the wall together.

→ **Joints:** Vertical joints should be staggered (separated) to prevent continuous lines of weakness.

5. Openings (Doors and Windows)

Openings weaken the structural integrity of walls:

→ A high number of door and window openings should be avoided.

→ Openings should not be constructed adjacent to the corners of the wall.

→ The total area of openings should be limited to **1/3rd** of the total length of the wall.

6. Reinforcement Detailing Provisions and bands

→ Enforce strong column-weak beam design.

→ Create boxing effect (i) vertical reinforcement (ii) Horizontal band (iii) Diagonal bracing (iv) Lateral restraints.

→ **Structural Bands:** These are essential for "box action" helping to distribute and transmit loads. Key bands include:

- o **Sill Band:** Below the windows.
- o **Lintel Band:** Above doors and windows.
- o **Roof/Gable Band:** At the roof or gable level.

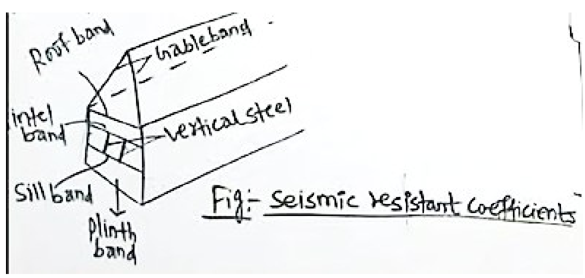


Figure: Creating Box Effect

→ Provide special confinement reinforcement (closely spaced ties) in beam-column joints and column ends.

→ Use high-ductility steel (Fe 550 or higher) and design-mix concrete (≥ M25).

→ Consider base isolation or dampers for critical or tall buildings

7. Parapet Height: The height of the parapet above the chajja (overhang) or balcony should be limited to **1 meter**.

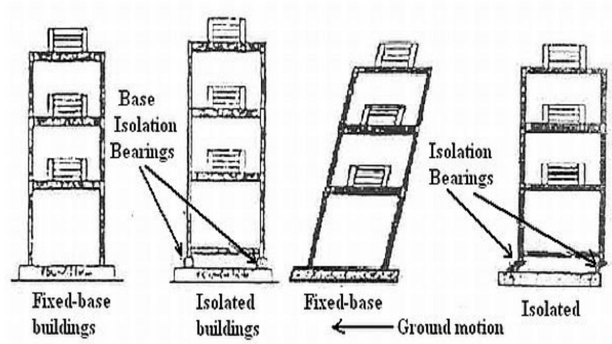
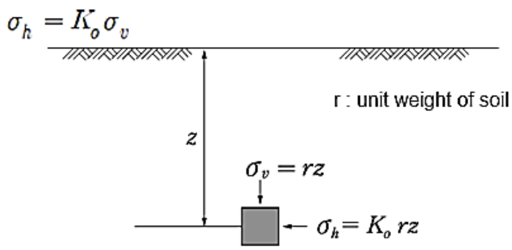


Figure: Base Isolation

3. What do you mean by effective earth pressure? What are the determining factors that affect the effective earth pressure of soil? [10]

Effective Earth Pressure:

The soil exerts vertical stress on underlying layers; some fractions of this vertical stress acts horizontally, thus horizontal exerted force due to effective stress (i.e. stress transmitted through point of contact of soil) not pore water pressure is called **effective earth pressure**.



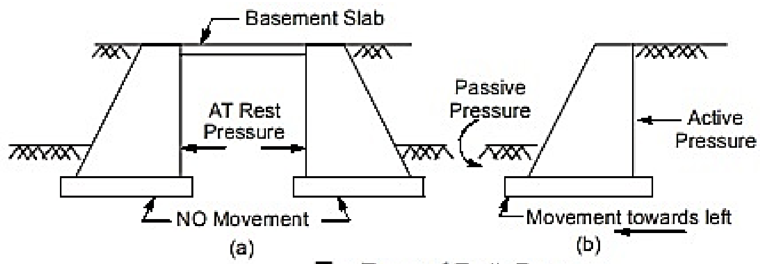


Fig: Types of Earth Pressure

Earth pressure is the pressure exerted by soil on a retaining wall. It is distinct from effective stress; while effective stress describes the compactness of soil grains, earth pressure specifically describes the lateral pressure against a structure.

Vertical Effective Stress: $\sigma'_v = \gamma H$ (where H is the height of soil).

Effective Lateral Earth Pressure: $\sigma'_h = K\sigma'_v = K\gamma H$ (where K is the coefficient of earth pressure).

There are mainly three types of Earth pressure based on the movement of the retaining wall:

Type	**Wall Movement**
Active Earth Pressure (P_a)	Wall tends to move **away** from the backfill.
Passive Earth Pressure (P_p)	Wall tends to move **into** the backfill.
At Rest Pressure (P_0)	**No movement** of the retaining wall.

Factors Affecting Earth Pressure

A. Properties of Soil

The unit weight (γ) and angle of internal friction (ϕ) affect pressure.

Active Earth Pressure Force: $P_a = \frac{1}{2} K_a \gamma H^2$

Rankine's Coefficient (K_a):

$$K_a = \frac{1 - \sin\phi}{1 + \sin\phi}$$

B. Water Table

The presence of water modifies the unit weight and adds pore water pressure (u). Lowering the water table increases effective stress, which in turn increases earth pressure.

C. Surcharge (q)

Surcharge refers to an overburden weight or load applied on the soil surface. This increases the vertical effective stress.

Total Stress (σ): $\gamma H + q$

Pore Water Pressure (u): 0 (if dry)

Effective Stress (σ'): $\sigma - u = \gamma H + q$

Earth Pressure Formula (with Cohesion c): $P_a = K_a \sigma'_v - 2c\sqrt{K_a}$

D. Type of Soil

Earth pressure values differ significantly between **cohesionless** soil (sand) and **cohesive** soil (clay).

E. Relative Movement of the Wall

The magnitude and direction of the wall's movement relative to the backfill determine whether the pressure state is active, passive, or at rest.

Relationship between coefficients: $K_p > K_0 > K_a$

4. **What are the advantages of well foundation over the other foundations? Write short notes on methods and equipment of soil compaction in civil infrastructures construction.[5]**

The advantages of well foundation over the other foundation are:

→ Using well foundation desired depth of foundation can be achieved.
→ The effect of scour due to water on foundation can be better analyzed by using well foundation.
→ It is free from vibration, so easy and safe to construct.
→ Disturbance on surrounding structure is lowered by using well foundation.
→ It is used for the foundation construction of bridge, aqueducts etc.
→ It is better used when good soil is available only at greater depth.

Soil compaction is the mechanical process of increasing the density of soil by reducing air voids, that tends to increase bearing capacity, reduce settlement, and decrease permeability in civil infrastructure.

Field Methods of Soil Compaction & Their Suitability

It depends primarily on the **area** and the **type of soil**.

Manual and Small-Scale Methods

These methods are best suited for **narrow areas** where heavy machinery cannot maneuver.

- **Rammer (Hand Rammer):** A manual tool used for localized compaction.
- **Vibratory Plate Jumper:** A mechanical version of a rammer with a handle for better control.

Best for: Inside buildings, Behind retaining walls**,** Trenches and so on.

Large-Scale Methods (Rollers)

For **widely spread areas** such as roads, runways, and dams, different types of rollers are used based on the soil composition.

General Applications for Large Equipment

Roller Type	Ideal For	Specific Notes
Steel Drum Roller	Asphalt & Coarse-grained soil	Used for surface dressing and smooth finishing.
Pneumatic Tyred Roller	Coarse-grained soil	Best when fines (silt/clay) are **less than 20%**.
Sheep Foot Roller	Cohesive Soil	Features "feet" that penetrate the soil; best when fines are **greater than 20%**.
Vibrating Roller	Sand	Uses vibrations to rearrange particles into a dense state.
Grid Roller	Gravel & Weathered Rock	Has a grid-like surface with holes to break up and compact stony material.

Section B

5. The anneal peak discharge of a river follows the Gumbel's extremes value distribution with mean of 10000m/s and a standard deviation of 3000 m³/s. What is the probability that the annual peak discharge is more than 15000 m³/s. What is the magnitude of the peak discharge with an exceedance probability of 0.1? [5]

Given:

Mean, $\bar{x} = 10000 \text{ m}^3/\text{s}$

Standard deviation, $\sigma = 3000 \text{ m}^3/\text{s}$

For a Gumbel distribution:

$$\text{Mean} = \mu + \gamma\beta$$
$$\sigma = \frac{\pi}{\sqrt{6}}\beta$$

where

$\gamma = 0.5772$(Euler's constant)

μ= location parameter

β= scale parameter

Step 1: Determine Gumbel parameters

Scale parameter β

$$\beta = \sigma\frac{\sqrt{6}}{\pi} = 3000 \times \frac{2.449}{3.142} \approx 2337 \text{ m}^3/\text{s}$$

Location parameter μ

$$\mu = \bar{x} - \gamma\beta = 10{,}000 - (0.5772)(2337) \approx 8650\text{m}^3/\text{s}$$

Probability that annual peak discharge exceeds 15,000 m³/s

CDF of Gumbel distribution:

$$F(x) = \exp\left[-\exp\left(-\frac{x-\mu}{\beta}\right)\right]$$
$$P(X > x) = 1 - F(x)$$

For $x = 15{,}000$:

$$z = \frac{15000 - 8650}{2337} \approx 2.72$$
$$F(15000) = \exp[-\exp(-2.72)] = \exp(-0.066) \approx 0.936$$

Probability ≈ 6.4%

Peak discharge with exceedance probability = 0.1

Exceedance probability $= 0.1$

Non-exceedance probability $F(x) = 0.9$

$$0.9 = \exp\left[-\exp\left(\frac{x-\mu}{\beta}\right)\right]$$
$$\exp\left(-\frac{x-\mu}{\beta}\right) = -\ln(0.9) = 0.1053$$
$$\frac{x-\mu}{\beta} = 2.251$$
$$x = \mu + 2.251\beta$$
$$x = 8650 + (2.251)(2337) \approx 13{,}900 \text{ m}^3/\text{s}$$

6. **When was the first hydropower commissioned in Nepal? describe hydropower development history in Nepal. Give list of electrical and mechanical equipment commonly installed in hydropower plants. [5+5]**

First Plant: The first hydropower plant in Nepal is **Chandra Jyoti** (located in Pharping) with an installed capacity of **500 kW**. It was constructed on **1968 Jestha 9** (B.S.) with aid from the British government. This marked the inception of hydropower in Nepal.

→ After Pharping, the **Sundarijal** hydropower plant was constructed.

→ The Energy Development Corporation (EDC) was established in 2019 to develop hydroelectricity.

→ Research indicates Nepal has a potential of **83,000 MW** from its abundant water sources, of which **43,000 MW** is technically feasible. As of now, **3336 MW** of hydroelectricity is being generated.

Notable Projects: Kaligandaki A, Trishuli, Devighat, Panauti, and Khimti have been constructed across different timeframes.

Policies & Regulations

Electricity Act 2042: Commissioned to regulate hydropower development.

Nepal Electricity Authority (NEA): Established under the NEA Act. NEA Act 2048 & Regulation 2050: Established to ensure the involvement of the private sector in Generation, Transmission, and Distribution (GTD).

Trade: Trans-boundary trade has been commissioned to treat electricity as an exportable commodity.

Equipment's required in a Hydropower Plant

Electrical Equipment

Generator: Converts rotational energy into electrical energy.

Transformer: Steps up the voltage (increases it) at the power plant and steps it down at substations for distribution.

Switchgear: Used for controlling and protecting the electrical circuit.

Fuse: A safety instrument consisting of a thin metal wire that melts when high voltage/current passes through it to prevent electrical hazards.

Electrical Poles and Wires: Used for transmission and distribution.

Rotating (Mechanical) Equipment

Turbine: Converts the potential energy of water into rotational energy.

Governors: Regulate the rate of water flow when the output changes.

Shafts: Coupled with the turbines to transfer energy to the generator.

Bearings: Support the rotating shafts.

7. **What is rainwater harvesting? Although Nepal has the second-largest per-capita water availability in the world, the area of uncultivated barren land is increasing every year. Analyze the reasons behind this situation. As an engineer working for the Government of Nepal (GoN), propose practical measures to address this problem. [10]**

Rainwater harvesting is the process of collecting and storing rainwater (either underground, at the surface, or below ground) for use during water scarcity.

It is a modern approach of conjunctive use of water and is generally applied in arid and semi-arid regions.

Process: Rainwater from roofs, yards, balconies, and pavements is collected and diverted through pipes to be stored in tanks or reservoirs for further use.

Reasons for Increasing Uncultivated Barren Land

→ **Reduction in Active Population:** Due to a reduction in the active population (primarily caused by foreign employment and other factors), interest in agriculture is greatly reduced. This lack of interest discourages farming and leads to an increase in uncultivated barren land.

→ **Scarcity of Water:** Due to a limited supply of irrigation water in the fields, the demand for water cannot be met. This water shortage makes farming difficult, leading to more land becoming barren.

→ **Reduction in Water Table:** Due to the reduction in the water table level (specifically mentioned in the **Terai region**), a sufficient crop yield cannot be achieved. This decline in productivity results in an increase in barren land.

→ **Productivity reduction of soil:** Due to overuse of chemical fertilizers

→ **Insufficient Irrigation Infrastructure:** Due to lack of capital budget and managerial activity in irrigation sector

Proposed Practical Measures (Engineering Perspective)

Practical measures an engineer can take to address irrigation and agricultural problems:

→ **Change in Irrigation Pattern:** To increase the efficiency of irrigation, not only surface irrigation but also **subsurface and lift irrigation** can be practiced.

→ **Crop Rotation:** Crop rotation enhances crop yield and productivity. There are some crops which can grow even when there is less water; this method reduces the requirement for standing irrigation water.

→ **Development of Irrigation Infrastructure:** Optimum development of irrigation infrastructure like **canals, regulators, and waterways**, and projecting new plans for irrigation development reflect the state's vision for agriculture. This encourages farmers and reduces the problem of barren land.

→ **Encouraging Cash Crop Production:** Testing the soil and deciding which plants can grow better in specific soil types also helps in reducing barren land.

→ **Legislative and Managerial Improvement:** The formation of **Water User Associations (WUA)** and **Farmer Managed Irrigation Systems (FMIS)**, along with a more pronounced volumetric assessment of water, can help to reduce barren land.

Section C

8. **Describe briefly the methodology used in the construction of gravel roads. Differentiate between surfaced dressing treatment and other seal construction. [5+5]**

A **Gravel Road** consists of broken stone aggregates which are placed and compacted, typically providing a **camber of 4%**. Gravel roads can carry a heavier wheel load than earthen roads.

Methods of Construction

There are generally two types of construction methods for gravel roads:

a) **Feather Type:** The gravel layer is thicker at the center and tapers off toward the edges.

b) **Trench Type:** A trench is dug into the subgrade, and the gravel is filled within it, maintaining a uniform depth or specific profile.

FIG. 3
TRENCH TYPE GRAVEL ROAD

FIG. 4
FEATHER EDGE TYPE GRAVEL ROAD

Construction Steps

The process follows a specific sequence to ensure durability and proper drainage:

→ **Subgrade Preparation:** The correct subgrade level is achieved. All loose soil is cleared, and the remaining soil is compacted to create a firm foundation.

→ **Stacking the Material:** Stacking the gravel along the proposed road alignment helps to reduce transportation costs and ensures materials are ready for use.

→ **Placing:** The gravel is placed in the trench (or on the subgrade). The thickness of the gravel is typically greater at the center than at the edges to assist with water runoff.

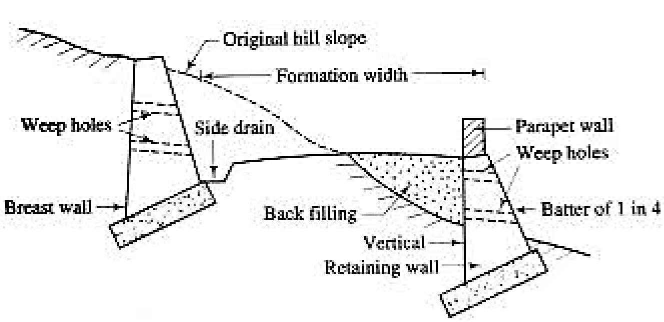
→ **Compaction:** The gravel is compacted using rollers. Compaction is done from the **center toward the edges** to achieve the desired camber (slope).

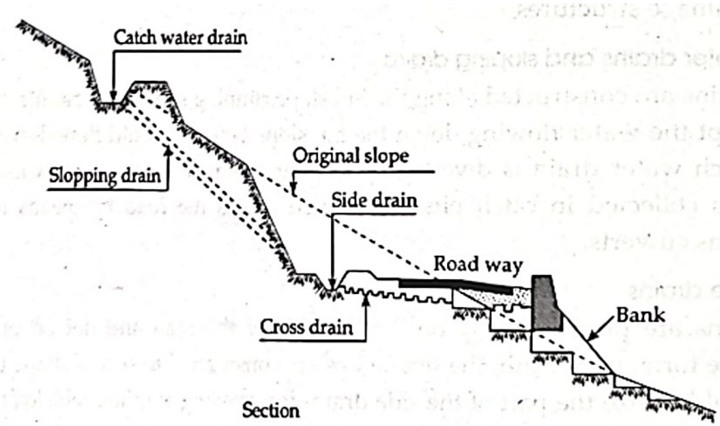
→ **Opening the Traffic:** Once compaction is complete and the surface is set, the road is opened for public use.

Comparison of surface dressing treatment and other seal construction

Feature	**Surface Dressing (Chip Seal)**	**Slurry Seal / Microsurfacing**	**Fog Seal**
Composition	Binder + Large Stone Chips	Pre-mixed Slurry (Fine Agg)	Diluted Bitumen Only
Application	Layer by layer (Spray then Chip)	Single pass (Mixed in machine)	Spray only
Surface Texture	Rough (High skid resistance)	Smooth (Like asphalt)	Existing texture remains
Main Purpose	Waterproofing & Skid Resistance	Filling minor voids & Aesthetics	Sealing & Rejuvenation
Traffic Suitability	High to Medium Volume	Low to Medium (Residential)	Low Volume / Shoulders

9. Explain the highway cross-section used in hilly roads. [10]





Highway cross-sections in hilly terrain require specialized designs to manage steep slopes (>25%), involving cut-and-fill, retaining walls, and robust drainage.
Key elements include narrow pavements with steep cambers (3%–4%), uphill-side catch-water drains, valley-side parapet walls, and hairpin bends.

Key Components of a Hilly Road Cross-Section

→ **Formation Width:** Often narrower than plain roads, consisting of carriageway and shoulders.
→ **Cut and Fill:** Roadways are often benched into the hillside, using material from the cut side (hill side) to fill the valley side.
→ **Drainage Structures:** Critical for stability, including:
  - **Side Drains:** Placed on the uphill side to carry surface runoff.
  - **Catch-Water Drains:** Located above the cutting to intercept water from the slope.
  - **Culverts:** Placed every 60–100 m in high-rainfall areas.

→ **Stability Structures:**
  - **Retaining Walls:** Constructed on the valley side to support the embankment.
  - **Breast Walls:** Built on the uphill side to prevent landslides and rockfalls.
  - **Parapet Walls:** Safety walls on the valley side to protect vehicles.

→ **Camper/Slope:** Typically, the road is sloped towards the side drain (inward drainage).

Key Types of Cross-Sections in Hilly Roads:

Cut and Fill: The most common method, where a portion of the hillside is excavated (cut) and used to create an embankment (fill) on the valley side, supported by a retaining wall.

Bench Type (Benching): The hillside is cut into a series of steps or "benches" to provide a stable foundation for the road embankment, reducing the risk of slope failure.

Box Cutting: A method used where the road is completely enclosed by cutting into the hill on both sides, often in steep terrain.

Semi-Bridge: When the steepness prevents a full-width road on solid ground, a portion of the road is supported by a retaining wall, and the other part may be supported by a bridge structure on the valley side.

Semi-Tunnel: Used in very steep rock faces where the road is cut into the rock, allowing the overhanging rock to form a natural roof, reducing the amount of excavation required.

10. What is SSD and OSD? How are these two terms related to each other? [5]

Stopping Sight Distance (SSD)

SSD is the minimum distance required for a driver traveling at design speed to bring their vehicle to a complete stop before reaching a stationary object on the road. It is the sum of two distances:

→ **Lag Distance:** The distance traveled during the driver's perception-reaction time.
→ **Braking Distance:** The distance traveled after the brakes are applied.

The formula for SSD is:

$$\mathrm{SSD} = \mathrm{v}\mathrm{t} + \frac{\mathrm{v}^2}{2\mathrm{g}(\mathrm{f} \pm \eta)}$$

Where:

v = speed of the vehicle
t = reaction time (usually 2.5 seconds)
g = acceleration due to gravity
f = coefficient of friction
η = gradient of the road

2. Overtaking Sight Distance (OSD)

OSD (also known as Passing Sight Distance) is the minimum distance open to the vision of a driver of a vehicle intending to overtake a slow vehicle ahead with safety against traffic in the opposite direction. It includes:

→ The distance traveled during perception time. (d_1)

→ The distance traveled during the actual overtaking maneuver. (d_2)

→ The distance traveled by the oncoming vehicle during the maneuver. (d_3)

The primary formula combines these: OSD = $d_1 + d_2 + d_3$

Where, $d_1 = v_b \times t$; $d_2 = v_b \times T$ and $d_3 = v \times T$

Key Components (IRC Method)

v_b: Speed of the slow-moving (overtaken) vehicle (m/s).

v: Speed of the overtaking vehicle (m/s).

t: Driver's reaction time (typically 2 seconds).

T: Time taken to complete the overtaking maneuver.

s: Spacing between vehicles (e.g., $s = 0.69v_b + 6.1$ where, $v_b$in m/s).

How they are related:

→ **Design Priority:** SSD is a fundamental requirement for all highway sections, while OSD is needed only where overtaking is permitted. However, on horizontal and vertical curves where OSD is too expensive or difficult to maintain, provide **Intermediate Sight Distance (ISD)**, which is typically defined as 2 × SSD.

→ **Relative Magnitude** – OSD is significantly longer than SSD (usually 3 to 5 times longer) because overtaking requires time in the opposing lane plus a safety margin against oncoming traffic.

Section D

11. What is mean by treatment of water and why is it necessary? What do you mean dry flocculation? [5]

Water treatment is the process used to kill harmful pathogenic bacteria present in water and make it fit for domestic and industrial purposes.

Necessity of Water Treatment

The primary reasons for treating water include:

→ **Safety:** To kill harmful pathogenic bacteria, and microorganisms present in the water.

→ **Disease Prevention:** To reduce epidemics caused by contaminated water and minimize water-related diseases.

→ **Standards:** To supply water that meets minimum water supply quality standards.

→ **Public Health:** To reduce health risks for humans, particularly women and children.

→ **Socioeconomic Impact:** To improve the socioeconomic conditions of the people and the country.

Dry Flocculation

It is the process of adding a **coagulant** to water to remove fine suspended impurities that cannot be settled in a plain sedimentation tank.

Mechanism: When a coagulant is added to water, it combines fine sediments to create a **floc**. Because the floc is heavier than water, it can be settled at the bottom of the tank with the help of gravity. **Alum** (the most commonly used coagulant, known for forming good flocs).

12. **Compare the separate and combined sewer system. What is self-cleaning velocity and non-scouring velocity in sewer? Why are sewers not designed to full flow? [4+3+3]**

Separate vs. Combined Sewer Systems:

Feature	**Separate Sewer System**	**Combined Sewer System**
Function	Transports sanitary sewage (wastewater) and stormwater runoff in separate pipes.	Transports both sanitary sewage and stormwater runoff in a single pipe.
Advantages	-More efficient wastewater treatment -Reduced risk of overflows and backflow -Lower risk of environmental pollution	- Lower initial construction cost (one set of pipes)
Disadvantages	- Higher initial construction cost (two sets of pipes) - Requires more maintenance for two systems	- Increased risk of overflows during heavy rainfall, polluting waterways - Higher wastewater treatment costs during overflows due to diluted sewage
Suitability	- Densely populated areas - Areas with separate storm drain systems	- Older, established cities - Areas with limited space for separate pipes

Self-Cleaning Velocity: This is the minimum water flow velocity required to transport solids through the sewer pipe and prevent them from settling and accumulating.

Non-Scouring Velocity: This is the maximum water flow velocity that can safely travel through the sewer pipe without causing erosion or damage to the pipe lining.

Sewer systems are not designed to operate at full capacity for several reasons:

To Ventilate Sewer Gases: A space at the top allows harmful and explosive gases like methane and hydrogen sulfide to escape through designated vents, preventing dangerous build-up.

Accommodating Peak Flows: Sewer usage fluctuates throughout the day, with peak periods during showers, laundry cycles, etc. Designing for full flow would be wasteful for most of the day.

Preventing Surcharges and backflow: Full pipes can lead to backflow into homes and streets (sewer surcharges).

Optimal Hydraulic Efficiency: According to hydraulic principles, a circular pipe reaches its maximum discharge when it is approximately 93% full, and its maximum velocity at 81% full.

13. **Discuss in briefly how global warming is affecting water supply sector in Nepal and what will be the proper remedial measures? [5]**

Global warming is the rise in the average temperature of the Earth due to an increase in Greenhouse Gases (GHG).

How it Occurs:

→ **Balance of Nature:** Earth naturally maintains a balance between **sources** (GHG emitted) and **sinks** (GHG absorbed).

→ **Human Impact:** Due to human activity, "sources" increase while "sinks" decrease. This causes Greenhouse Gases to increase and become trapped in the upper layer of the atmosphere.

→ **Heat Trapping:** This thick layer of GHG on the upper layer of the Earth's surface traps solar radiation within the atmosphere, causing the Earth's temperature to rise.

Effect of Global Warming on W/S Sector in Nepal

a) **Reduction in Source:** Water sources are being depleted (drying up) due to the rise in temperature.

b) **Melting of Ice:** The melting of ice in the Himalayan region affects reliable perennial (year-round) water sources.

c) **Dryness of Lakes:** High temperatures cause faster evaporation from lakes, leading to a drop in water levels.
d) **Change in Rainfall Pattern:** Global warming causes shifts in rainfall patterns, resulting in the unavailability of water in appropriate quantities and quality for use.

Remedial Measures

To prevent these effects, the following measures are suggested:

→ **Emission Reduction:** Reduce the emission of Greenhouse Gases.
→ **Legislation:** Implement strict laws regarding industrial emissions and air quality.
→ **Waste Management:** Enforce the "3Rs" policy: **Reduce, Reuse, and Recycle**.
→ **Water Management:** Focus on the rational utilization of water resources.
→ **Afforestation:** Plant more trees to increase natural carbon sinks.
→ **Technical Solutions:** Implement deep lake cooling systems. Use **evaporetardars** to reduce water loss from surfaces.

THE END

MODEL SET 14

लोक सेवा आयोग

नेपाल इन्जिनियरिङ सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह,

राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

समय: ३ घण्टा **पूर्णाङ्क: १००**

पत्र: द्वितीय **विषय: Technical Subject**

तलका प्रश्नहरुको उत्तर Section अनुसार छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ ।

Section A

1. Write on design consideration of one-way slab. [5]

Here are the key design considerations for a one-way slab:

1. General Geometry and Depth

→ The slab is considered as a **one meter wide beam** for design purposes.

→ **Span to Depth Ratios: (from codal provision)**
- For a **simply supported**, l/d should be up to **30**.
- For **continuously supported**, l/d should be up to **35**.

2. Reinforcement Specifications

→ **Main Steel is** Provided in the **shorter direction**.

→ **Distribution Steel is** placed in the **longer direction** and positioned **above** the main steel.

→ **Maximum Diameter:** The maximum diameter of the main steel bar is $1/8^{th}$ of the overall depth of the slab.

→ **Clear Cover:** Minimum of **15 mm** or the maximum diameter of the bar (whichever is greater).

→ **End Cover:** The cover at the end of the reinforcing bar is 2ϕ or **25 mm** (whichever is greater).

Minimum and Maximum Reinforcement

Minimum Steel Area

→ **Mild Steel (Plain bars):** 0.15% of gross area.

→ **HYSD Bar (Fe 415/Deformed bars):** 0.12% of gross area.

Maximum Spacing (Center-to-Center)

The maximum spacing (S_{max}) between bars is determined by the effective depth (d):

For Main Steel: Lesser of 3d or **300 mm**.

For Distribution Steel: Lesser of 5d or **450 mm**.

Minimum Clear Spacing

The minimum clear spacing between bars must be the **maximum** of:

→ Nominal maximum size of coarse aggregate + 5 mm.

→ Maximum diameter of the bar (ϕ_{max}).

2. Nepal lies in earthquake prone zone, and we have faced severe earthquake recently. So, what do you suggest in designing and construction of earthquake resisting buildings in urban areas? [10]

Due to massive disaster from the 2015 Gorkha earthquake, all designs must adhere to the latest **NBC 105:2020** (Seismic Design of Buildings in Nepal), which provides following essential suggestions:

1. Appropriate Planning and Layout:

Prefer simple, symmetric building plans (e.g., rectangular shapes) to avoid torsional effects. Building with reentrant corner like U, L, V, H shaped in plan suffer more damage during earthquake.

2. Structural System Choice:

→ For mid-rise to high rise buildings, use reinforced concrete (RC) frames with shear walls for balanced stiffness and ductility.

→ Retrofit existing soft-story buildings with bracing or shear walls.

→ For new construction, incorporate shear walls around staircases/lifts.

→ Consider base isolation or dampers for critical or tall buildings.

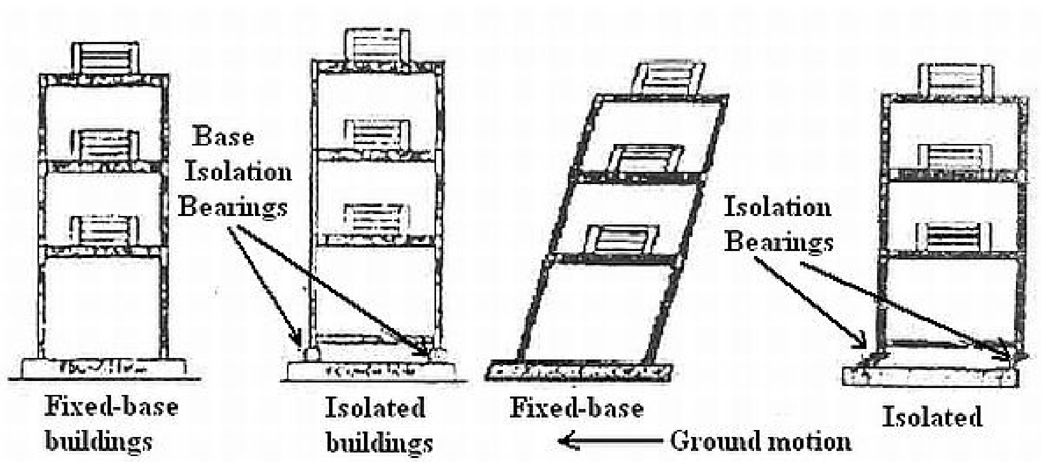


Figure 1: Base Isolation

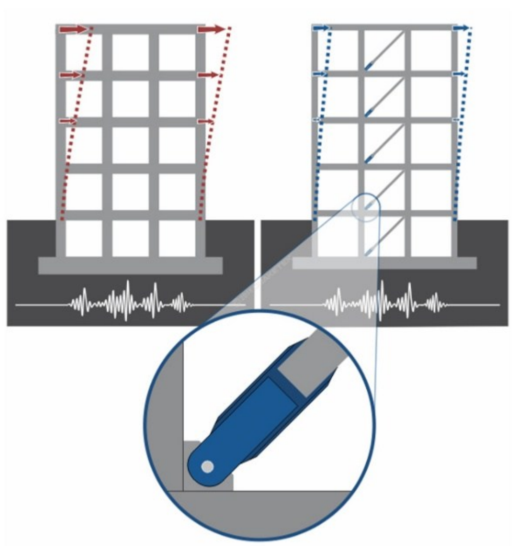


Figure 2: Seismic Dampers device

3. Reinforcement Detailing Provisions:

→ Enforce strong column-weak beam design.

→ Create boxing effect (i) vertical reinforcement (ii) Horizontal band (iii) Diagonal bracing (iv) Lateral restraints.

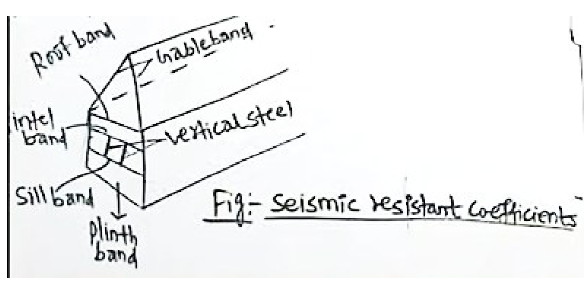


Figure 3: Creating Box Effect

→ Provide special confinement reinforcement (closely spaced ties) in beam-column joints and column ends.

→ Use high-ductility steel (Fe 550 or higher) and design-mix concrete (≥ M25).

4. Foundation & Soil Considerations:

In areas like Kathmandu Valley's soft soil:

→ Conduct mandatory geotechnical investigation.

→ Use raft or piled foundations in poor soil and address liquefaction risk.

→ Connect all footings with plinth beams.

5. Construction Quality Control:

→ Use certified materials and enforce proper curing of concrete.

→ Ensure correct placement of reinforcement especially closed stirrups and lap splices.

→ Daily supervision by qualified engineers is essential.

6. Non-Structural Element Safety:

Secure water tanks, parapets, cladding, and partitions with proper anchorage to prevent life hazards during shaking.

3. **What is flow net? Write its uses. [5]**

A **flow net** is a graphical representation used in seepage analysis, formed by the intersection of **flow lines** and **equipotential lines**. It provides a visual and mathematical way to study the flow of water through porous media (like soil) under structures like dams.

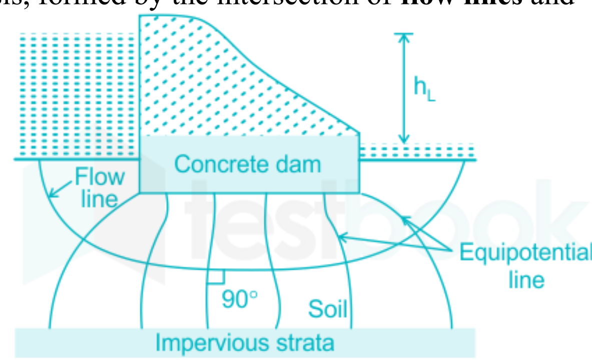


Figure 4: Flow nets

Key Components

→ **Flow Line:** Represents the path traced by an individual water particle as it moves through the soil.

→ **Equipotential Line:** A contour or line joining points of equal potential or hydraulic head.

Applications (Uses) of Flow Nets

Flow nets are generally used for determining the following four factors:

1. Quantity of Seepage (q)

The flow net allows for the calculation of the total amount of water leaking under or through a structure. It is calculated using the formula:

$$q = k \cdot h \left(\frac{N_f}{N_d}\right)$$

Where:

k = Coefficient of permeability

h = Total hydraulic head (difference in water levels)

N_f = Number of flow channels

N_d = Number of potential drops

2. Seepage Pressure (P_s)

It is used to determine the pressure exerted by the flowing water on the soil particles at any given point:

$$P_s = h_p \cdot \rho_w$$

(Where h_p is the pressure head and ρ_w is the unit weight of water).

3. Hydrostatic Pressure (u)

Flow nets help find the pore water pressure at a specific point:

$$u = (h_p)_p \cdot \gamma_w$$

4. Exit Gradient (i)

This is critical to prevent "piping" (soil erosion) at the downstream end. It is determined by the potential drop (Δh) over the length of the last field (Δs):

$$i = \frac{\Delta h}{\Delta s} = \frac{h/N_d}{\Delta s}$$

4. **What are the main types of foundation used for bridges? Describe in detail about underpinning with an aim to stabilize foundations of a bridge? [5+5]**

Bridge foundations are categorized as either **shallow** or **deep**, based on the depth of the load transfer to the subsoil.

A) Shallow Foundation

Open Foundation: Also known as a spread foundation, an RCC slab supporting the pier or abutment.
Soil Type: Suitable for strong soil layers, dense to very dense sand, rock layers, or boulder layers.
Structure: Ideal for **short span bridges**.
Location: Commonly used in **hilly areas**.

B) Deep Foundation

Parameters	**Pile**	**Pier**	**Caisson**
Size (diameter)	$\leq$ 2m($\leq$ 1.2m *mostly*)	2-3.5m($\approx$ 2m *mostly*)	$>$ 3.5m($>$ 5m *mostly*)
Body	Solid	Solid	Hollow
Layout pattern	m x n in group	Single or twin	Single (mostly)
Construction Method	Precast or cast-in-situ	Cast in situ	Casting—dredging—sinking
Depth	Best $\approx$ 20m	Best up to 40m	Best (20-30)m
Use soil	Clay or clayey soil	Clay extended to large depth	Sand or sandy soil
Structure	High rise building (Long span bridge)	Bridge	Bridge
Location	Ktm / Terai	Ktm	Terai

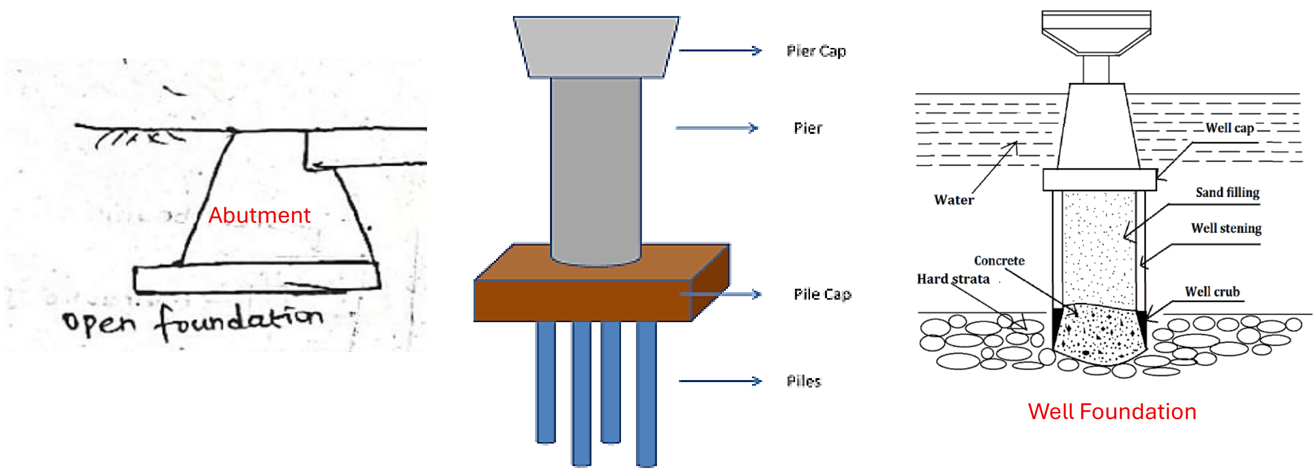


Figure 5: Types of foundations in Bridge

Underpinning is defined as the process of replacing the foundation for an existing structure with a larger or deeper type of foundation.

Objectives of Underpinning

→ **Deepening the Foundation:** To reach more stable soil strata or bedrock if the upper layers have shifted.
→ **Load Redistribution:** To allow the bridge to carry heavier modern traffic loads than originally designed for.
→ **Scour Protection:** To repair damage caused by river currents washing away soil from around bridge piers.

Methods of Underpinning

There are two primary methods:

1. Mass Concrete Underpinning (Pit Method)

This method involves digging pits underneath the existing foundation to reach more stable soil or to install a new foundation base.

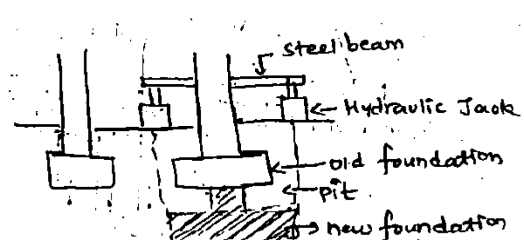


Figure 6: Pit Method

Steps involved:

Provide Support: Temporary support is provided for the wall or column.
Excavation: A pit is excavated beneath the existing structure.
Dismantling: The old foundation is dismantled.
Casting: Reinforcement is placed, and the new foundation is cast.

2. Pile Method (Micropiles/Jacked Piles)
The pile method uses long, slender columns (piles) to transfer the load of the building to a deeper, stronger soil layer.
Steps involved:
Construction: Piles are constructed on either side of the wall.
Insertion: A steel beam is inserted through the wall or column.
Support: The beam is rested on the piles to transfer the load, often using hydraulic jacks for precise weight distribution.

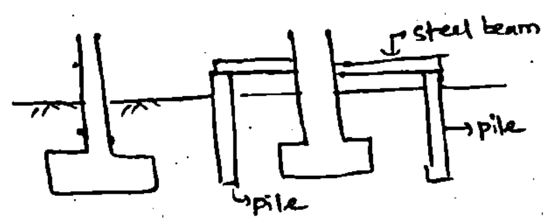


Figure 7: Pile Method

Section B

5. **What is boundary layer? Describe the term Displacement thickness, momentum thickness, and energy thickness. [5]**

When a solid body is immersed in a flowing fluid, there is a narrow region of the fluid in the neighbourhood of the solid body where velocity of fluid varies from zero to free stream velocity. This narrow region of fluid is called boundary layer in which there exists a large velocity gradient $\frac{dv}{dy}$ normal to boundary surface.

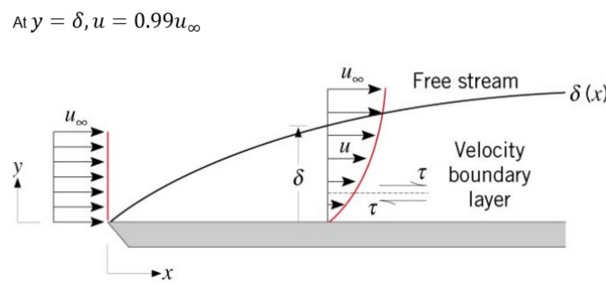


Figure 8: Boundary Layer

Displacement Thickness ($\boldsymbol{\delta^*}$)
It i1s defined as the distance measured perpendicular to the boundary by which the main/free stream is displaced on account of the formation of the boundary layer.

$$\delta^* = \int_0^\delta \left(1 - \frac{u}{U}\right) dy$$

Momentum Thickness ($\boldsymbol{\theta}$)
It is defined as the distance measured perpendicular to the boundary of the solid body, by which the boundary should be displaced to compensate for the reduction in **momentum** of the flowing fluid on account of boundary layer formation.

$$\theta = \int_0^\delta \frac{u}{U}\left(1 - \frac{u}{U}\right) dy$$

Energy Thickness ($\boldsymbol{\delta_e}$)
Energy thickness is defined as the distance measured perpendicular to the boundary of the solid body by which the boundary should be displaced to compensate for the reduction in **kinetic energy (K.E.)** of the flowing fluid on account of boundary layer formation.

$$\delta_e = \int_0^\delta \frac{u}{U}\left(1 - \frac{u^2}{U^2}\right) dy$$

6. **a) Write short notes on:**
i) Surface drainage and its application
Surface drainage is the orderly removal of excess water from the surface of land through improved natural channels or constructed ditches and through shaping of the land surface.
Applications:
→ Draining large quantities of surface water off land quickly.

→ Intercepting run-off water from resources (e.g., roadside drainage, neighboring properties, dam overflow).
→ Acting as a collector system from sub-surface drainage systems.

ii) Sub-surface drainage and its application
The process of removing excess water away from the root zones of plants by natural or artificial means, such as by using a system of pipes and drains placed below ground surface level, is called sub-surface drainage.
Applications:
→ Prevents water overflow.
→ Decrease the possibility of toxic materials contaminating our water supply.
→ Causes water to flow readily from land to ditch without harmful erosion or deposition of silt.

iii) Drainage coefficient (D.C)
The rate of water removal per unit area used in drainage design is known as the drainage coefficient.
→ It is expressed as the flow rate per unit area (i.e., $m^3/s/m^2$ or L/s/ha).
→ The drainage coefficient for agricultural land is decided such that no appreciable damage is caused to the crops to be grown in that land.
→ The intensity of rain and its duration are **inversely proportional** to the time allowed for the removal of water.

b) determine the size at the outlet of a 5 hectare drainage system, if the DC is 1 cm and tile grade is 0.03%, Assumes rigidity coefficient of the materials is 0.011. [5+5]
Given
Drainage area, $A = 5\text{ha} = 50{,}000\ \text{m}^2$
Drainage coefficient, $DC = 1\ \text{cm/day} = 0.01\ \text{m/day}$
Tile grade (slope), $S = 0.03\% = 0.0003$
Rigidity coefficient (Manning's n) $= 0.011$

1. Design discharge at the outlet

$$Q = \frac{A \times DC}{\text{time}}$$
$$Q = \frac{50{,}000 \times 0.01}{86{,}400} = \frac{500}{86{,}400} = 0.0058\ \text{m}^3/\text{s}$$

2. Use Manning's equation (pipe flowing full)

$$Q = \frac{1}{n} A R^{2/3} S^{1/2}$$

For a circular pipe flowing full:

$$A = \frac{\pi D^2}{4} \text{ and } R = \frac{D}{4}$$

Substituting and solving for D:

$$0.0058 = \frac{1}{0.011}\left(\frac{\pi D^2}{4}\right)\left(\frac{D}{4}\right)^{2/3}(0.0003)^{1/2}$$

Solving gives: $D \approx 0.19$ m
Required outlet drain size ≈ 0.19 m (≈ 190 mm diameter)

7. **Assume that you are assigned with responsibility of investigating a case of failure of a bridge immediately after its construction. What steps would you follow to investigate whether hydrology is a potential cause of the failure? Illustrate the steps with appropriate examples. Also, show a template of report (with notes on expected contents under different chapters/sub-chapters) of such investigation to submit to a higher authority.**
Investigation of Bridge Failure: Hydrological Perspective

To investigate if hydrology caused a bridge failure, follow these four critical steps based on the standard report framework:

1. Data Validation & Return Period Check
Re-evaluate the **"Reference hydrology data"** to see if the design flood was underestimated. Compare the flood that caused the failure against the design return period (usually 100 years in Nepal). If the actual flow exceeded the **"Design Flood"**, the failure may be due to an unprecedented natural event.

2. Peak Flow & Inflow Re-estimation
Recalculate the **"Estimation of inflows"** using different methods (e.g., Rational Method or WECS/DHM).
Example: If the bridge was designed using regional equations that don't fit that specific catchment's **"Biophysical characteristics,"** the peak flow capacity would be insufficient.

3. Scour and Morphological Analysis
Calculate the **Maximum Scour Depth** using the actual flood discharge. If the piers were founded above this depth because the "Silt Factor" was guessed incorrectly, the foundation would have been undermined.

Mean scour depth: $d_{sm} = 1.34 \left(\frac{Q_b^2}{K_{eff}}\right)^{1/3}$
K_{eff} (Lacey's factor): $K_{eff} = 1.76 \sqrt{d_{mm}}$
Maximum scour depth at pier: $d_{max} = 2 \times d_{sm}$
Maximum scour depth at abutment: $d_{max} = 1.27 \times d_{sm}$
Minimum depth of foundation below HFL: Minimum foundation depth $=$ Maximum scour depth $+$ Grip length

4. Hydraulic Capacity & Waterway Review
If the bridge opening (waterway) was too narrow, it would cause high-velocity flow and debris accumulation, leading to structural collapse or outflanking.

Part 2: Investigation Report Template

Chapter	Sub-Chapter	Expected Contents / Notes
1. Introduction	1.1 Objectives	To identify if hydrological design exceeded physical limits.
	1.2 Methodology	Data collection, hydraulic modeling, and site visit logs.
2. Watershed Status	2.1 Topography	Map of the catchment area contributing to the bridge site.
	2.2 Land Use Change	Identification of any recent changes (e.g., landslides) affecting runoff.
3. Hydromet Data	3.1 Data Validation	Comparison of design rainfall vs. actual rainfall during the failure event.
4. Inflow Re-estimation	4.1 Peak Flow Analysis	Recalculation of the hydrograph for the day of failure.
	4.2 Flow Duration	Comparing the event flow against the "Flow Duration Curve."
5. Design Flood Review	5.1 PMP/PMF Analysis	Was the "Probable Maximum Flood" considered for the safety of the structure?
	5.2 Frequency Check	Determining the return period of the actual failure event.

6. Hydraulic/Sediment	6.1 Morphological Impact	Evidence of bed scouring or debris blockage (from "Sedimentation Studies").
7. GLOF/Climate	7.1 Extreme Events	Investigation into Glacial Lake Outbursts (GLOF) if in a high-altitude zone.
8. Conclusion	8.1 Causality	Clear statement on whether hydrology was the primary cause (e.g., "Design flow was underestimated by 30%").

Section C

8. A highway curve with a radius of 500 meters is designed for a design speed of 80 km/h. Calculate the required superelevation (banking) of the curve, assuming a coefficient of lateral friction (friction between tires and road surface) of 0.15. [10]

Given

Radius of curve, ($R = 500$ m)

Design speed, ($V = 80\ \mathrm{km/h} = 22.22\ \mathrm{m/s}$)

Coefficient of lateral friction, ($f = 0.15$)

Acceleration due to gravity, ($g = 9.81, \mathrm{m/s^2}$)

For a horizontal curve:

$$e + f = \frac{V^2}{gR}$$

Substitute values:

$$e + 0.15 = \frac{(22.22)^2}{9.81 \times 500}$$

$$e + 0.15 = \frac{493.8}{4905} \approx 0.1007$$

$$e = 0.1007 - 0.15 = -0.0493$$

Hence, The negative value means **friction alone is more than sufficient** to handle the centrifugal force at 80 km/h on a 500 m radius curve.

Required superelevation: $e = 0$; (no banking required)

9. Why is road maintenance necessary? Describe different types of road maintenance. Explain the maintenance of the bituminous pavement. [10]

Need for Maintenance:

(i) To preserve the road asset and to deliver the designated level of service to the road users.

(ii) Rs. one investment in maintenance saves Rs. 3 to 6 required later for rehabilitation and reconstruction.

Based on the handwritten notes in the image, here is the transcribed and structured answer to the question regarding road maintenance.

Classification of Road Maintenance in Nepal

According to the **Department of Roads (DoR)** and the **Road Board Nepal (RBN)**, maintenance activities are classified into five main categories:

1) **Routine Maintenance:** Small-scale, continuous activities carried out year-round, such as cleaning drains, cutting grass, and clearing minor landslides.
2) **Recurrent Maintenance:** Periodic activities required a few times a year, such as pothole patching, crack sealing, and edge repair.
3) **Periodic Maintenance:** Major scheduled works carried out every few years (e.g., 5–7 years), such as applying a new **Seal Coat** or an asphalt overlay to restore the riding surface.

4) **Emergency Maintenance:** Immediate repairs required due to sudden failures, often caused by heavy monsoon rains, large landslides, or floods, to restore road connectivity.
5) **Specific Maintenance:** One-off repairs like building a new retaining wall or replacing a damaged culvert that does not fall under routine or periodically.

Maintenance of bituminous roads involves timely, systematic repairs to ensure structural integrity and safety. It prevents the deterioration of the road surface and extends its service life.

Types of Maintenance for Bituminous Pavement

1. Patch Repairs

This is the process used to fix potholes and localized failures.

→ Cut damaged areas into rectangular shapes, remove debris, and clean up to ensure the hole is dry.
→ Apply a tack coat of emulsion or cutback bitumen to the sides and bottom.
→ Fill the potholes with pre-mix and compact thoroughly using a roller or hand rammer to achieve a dense surface.

2. Crack Sealing & Filling

Cracks can lead to significant structural issues if water penetrates the road layers.

→ Clean the cracks and then apply sealants or bituminous material to seal the opening.
→ This helps to prevent water infiltration, which weakens the subgrade.

3. Edge Repair

The edges of the road are often susceptible to wear and tear from heavy traffic or lack of support.

→ Fixing broken or raveled edges to prevent shoulder failure and maintain the width of the pavement.

4. Shoving Repair

Shoving refers to the formation of ripples or "waves" on the road surface, usually where vehicles brake or accelerate.

→ Correcting depressions and waves by milling (scraping off high spots) or adding material to restore a smooth profile.

5. Drainage Maintenance

Regular cleaning of side ditches and culverts to ensure proper water runoff, preventing waterlogging of the bituminous layers.

10. What factors to be considered when selecting a location for airport? List down the factors. [5]

Factors that should be considered during airport site selection are as follows:

→ **Regional plan:** Alignment with the broader development goals of the area.
→ **Airport use:** Determining whether the airport is for commercial, military, or private use.
→ **Proximity to other airports:** Ensuring there is no airspace conflict or redundant service.
→ **Ground accessibility:** Ease of transport for passengers and cargo to and from the city.
→ **Topography:** The physical features and "lay of the land."
→ **Obstructions:** Identifying natural or man-made hazards like mountains or tall buildings.
→ **Visibility:** Assessing prevalence of fog, smoke, or haze that could affect flight operations.
→ **Wind:** Direction and intensity, which are critical for runway orientation.
→ **Noise nuisance:** Impact on surrounding residential or sensitive areas.
→ **Grading, drainage, and soil characteristics:** Technical requirements for stable construction and water management.
→ **Future development:** Potential for expanding the airport as demand grows.
→ **Availability of utilities from town:** Access to water, electricity, and telecommunications.
→ **Economic considerations:** Cost-benefit analysis, including land acquisition and construction expenses.

Section D

11. Why aeration in used in water treatment plants? Is it more commonly made with ground water surface water and why? give reasons [5]

Aeration is the process of bringing water and air into close contact. This is done to remove dissolved gases (such as carbon dioxide) and to oxidize dissolved metals like iron. It can also be used to remove volatile organic chemicals from the water.

Aeration used in Water Treatment Plants because of the following reasons:

→ Oxidation of Organic Matter
→ Increase Dissolved Oxygen
→ **Taste and Odor Removal:** Reduces the concentration of substances like hydrogen sulfide (H_2S) and various organic compounds through volatilization or oxidation.
→ **Metal Removal:** It oxidizes dissolved iron and manganese, rendering them **insoluble** so they can be filtered out.
→ Aids in the flocculation of colloids in sewage influent.
→ **Efficiency:** Removes compounds that might interfere with or increase the cost of subsequent treatment steps.

Aeration is **more commonly used for groundwater** (from wells and boreholes) rather than surface water.

Why Groundwater?

→ Groundwater interacts with underground rocks, leading to high levels of dissolved gases like CO_2 and H_2S, as well as metals like iron and manganese.
→ **GW** is devoid of oxygen, so aeration is necessary to manually add oxygen to oxidize contaminants.

Why not Surface Water?

→ Surface water is already in contact with the atmosphere for prolonged periods.
→ It typically has sufficient natural oxygen levels and is less likely to have high concentrations of dissolved hydrogen sulfide or methane.

12. Explain Multiple tube fermentation technique. What are water related diseases? Describe briefly. [4+6]

Multiple Tube Fermentation Technique: It is a standard, three-stage microbiological method (presumptive, confirmed, completed) used to detect and enumerate coliform bacteria in water by measuring gas production in lactose broth. It estimates the Most Probable Number (MPN) of coliforms per 100 mL, providing a statistical density of bacteria.

1. Presumptive Test

→ Samples are taken into standard fermentation tubes called Durham tubes containing **Lauryl Tryptose Broth (LTB)**.
→ A positive result is indicated by **gas formation** within 24–48 hours at 35°C.
→ This indicates the potential presence of coliforms.

2. Confirmed Test

→ Samples from positive presumptive test tubes are transferred to **Brilliant Green Lactose Bile Broth (BGLB)**.
→ The tubes are incubated at $35\ \pm 0.5°C$ for 48 hours.
→ If gas is formed, it is considered a **confirmed positive** for coliform.

3. Completed Test

→ Samples from confirmed positive tubes is marked as streaks in the plates containing **Eosin Methylene Blue (EMB) agar**.
→ The plates/tubes are incubated at $35\ \pm 0.5°C$ for 24 hours.
→ Typical colonies are then transferred back to a broth and an agar slant to perform a **gas test** and **Gram staining**.
→ If gas formation is observed and Gram-stained samples are examined (confirming Gram-negative, non-spore-forming rods), it indicates **confirmed coliform presence**.

Classification of Water-Related Diseases

Water acts as a primary cause for various illnesses, which are classified into four main categories:

1. Water-Borne Diseases

These diseases arise from **infected water** and are transmitted when the water is used for **drinking or cooking**.

Examples: Diarrhoea, Typhoid, Paratyphoid, Dysentery, Hepatitis A, Cholera, and Jaundice.

Prevention: Arranging public awareness programs about wholesome water, improving water quality, and keeping water sources clean.

2. Water-Washed Diseases

These occur as a result of **inadequate quantities of water** being available for good personal hygiene.

Examples: Fungal skin diseases (like ringworm), ophthalmic diseases (eye diseases like trachoma and conjunctivitis), and infections caused by lice, mites, fleas, or ticks.

Prevention: Increasing the quantity of water available for personal use.

3. Water-Based Diseases

These are caused by **bacteria or parasites** that spend part of their life cycle in water. These organisms typically enter the human body through the **skin**.

Examples: Schistosomiasis and other parasite-related illnesses.

Growth: The growth of these bacteria occurs in water, and the disease spreads through the utilization of this water.

4. Water-Vector Diseases

These illnesses are transmitted by insects (such as mosquitoes or flies) that **breed in or rely on water**, even though the transmission isn't direct from the water itself.

Examples: Malaria, Sleeping Sickness, and Yellow Fever.

Prevention: Keeping surroundings clean and filling in trenches where water might collect and allow insects to breed.

13. What is environmental impact assessment? Write about its importance for environmental protection. [5]

EIA is a systematic approach used to identify, predict, and evaluate the biophysical, social, and other relevant effects and assess the environmental consequences of a project before it begins.

The **Environment Protection Act, 2076 (2019)** and **Environment Protection Rules, 2077 (2020)** provide the legal foundation for EIAs in Nepal.

Importance of Environmental Protection:

→ **Biodiversity Conservation:** EIA helps identify if a project will disrupt wildlife or endangered species, allowing for redesigns or "wildlife-friendly" infrastructure.

→ **Mitigation of Natural Hazards:** EIA assesses risks such as **Glacial Lake Outburst Floods (GLOFs)**, landslides, and soil erosion.

→ **Climate Change Resilience:** Newer EIA guidelines in Nepal encourage assessing how a project will contribute to or be affected by climate change, helping build "climate-smart" infrastructure.

→ **Social and Cultural Safeguards:** The EIA process includes mandatory **Public Hearings**, ensuring that the rights of indigenous communities and local people are protected and that their "National Heritage" sites are not destroyed.

→ **Legal Accountability:** Under the **EPA 2076**, implementing a project without an approved EIA or violating its terms can lead to heavy fines (up to **50 Lakh NPR**) and project suspension, acting as a warning against environmental negligence.

THE END

MODEL SET 15

लोक सेवा आयोग

नेपाल इन्जिनियरिङ सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह,

राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

समय: ३ घण्टा **पूर्णाङ्क: १००**

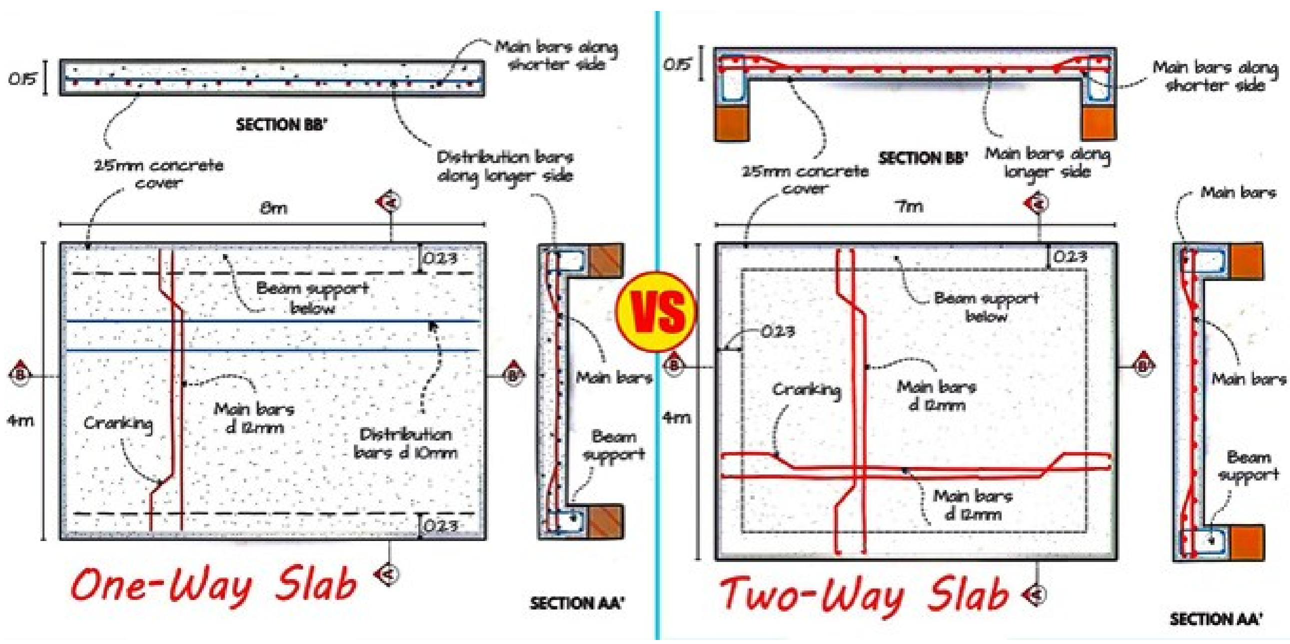
पत्र: द्वितीय **विषय: Technical Subject**

तलका प्रश्नहरुको उत्तर Section अनुसार छुट्टाछुट्टै उत्तरपुस्तिकामा लेख्नुपर्नेछ।

Section A

1. How do you differentiate between one way and two-way slab. [10]

One way slab	**Two-way slab**
i) Ratio of Longer/Shorter side ($L_y/L_x \geq 2$)	i) $L_y/L_x < 2$
ii) BM and deflection occur only in shorter span direction.	ii) BM and deflection occur in both the directions.
iii) Main reinforcement is provided only in shorter span and distribution reinforcement in longer span.	iii) Main reinforcement is provided in both directions.
iv) Deflection shape is cylindrical.	iv) Deflection shape is dish or sauce-shaped.
v) Crank bars are provided in shorter direction.	v) Crank bars are provided in both directions.
vi) Depth of slab is greater; amount of steel requirement is less.	vi) Depth of slab is lesser; amount of steel requirement is greater.
vii) Economical up to span of 3.6m.	vii) Economical up to grid 6m x 6m.
viii) Supporting beam provided on two longer sides.	viii) Supporting beams provided four sides.
ix) No Provision of torsional reinforcement	ix) Torsional reinforcement is needed for corner held down cases.
x) Load is carried by slab in one direction	x) Load are carried by slab in both directions
xi) Load transfer mechanism is triangular.	xi) Load transfer mechanism is triangular and trapezoidal.



2. **Draw a sketch of a T beam bridge and indicate its components.[5]**

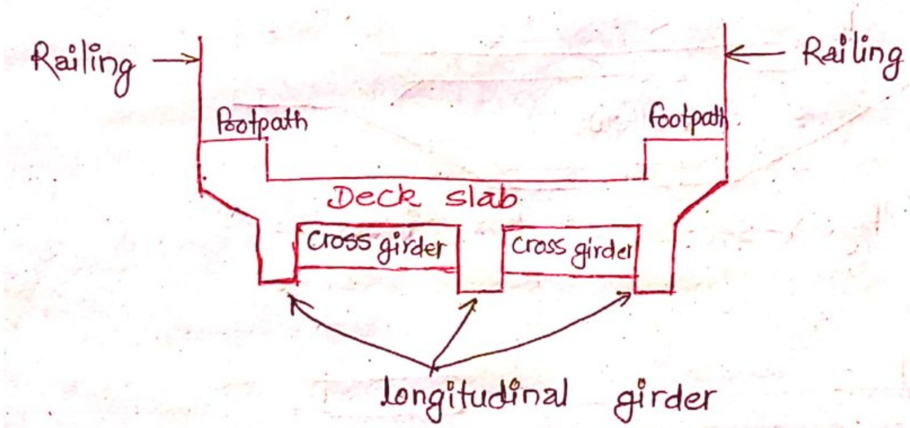


Fig 1: X-section of T-Beam Bridge

3. **Define and differentiate between compaction and consolidation of soil mass. Write down assumptions for Terzaghi's one-dimensional consolidation theory. [5+5]**

Compaction refers to the process of densification of soil mass by expulsion of air voids making soil particles closer with the help of mechanical means to improve its properties before it is put to any use.

Consolidation refers to the reduction in volume of soil due to expulsion of water from the voids.

The difference between compaction and consolidation are as follows: -

Compaction	Consolidation
i) It involves expulsion of air from voids.	i) It involves expulsion of water from voids.
ii) It is an artificial process.	ii) It is usually natural phenomenon.
iii) It mainly occurs in coarse-grained soils.	iii) It mainly occurs in fine-grained soils.
iv) No drainage required.	iv) Drainage of pore water is essential.
v) Proctor compaction test is performed for testing compaction.	v) Oedometer (consolidation) test.
vi) It is usually immediate process.	vi) Time-dependent (slow process).

Assumptions for Terzaghi's One-Dimensional Consolidation Theory:

1. The soil is homogeneous and isotropic.
2. Solid particles and water in voids are incompressible. Consolidation occurs solely due to the **expulsion of water** from the voids.
3. The soil is fully saturated.
4. The coefficient of permeability of the soil has the same value at all points and remains constant during the entire period of consolidation.
5. Darcy's law is valid (flow velocity is proportional to the hydraulic gradient).

4. **Illustrate the types of slope failures with suitable sketches. A vertical cut is made in a clay deposit. [consider: C=30 KN/m², ϕ= 0, γ = 16 kN/m³, Fc= 1.00 and Sn=0.261]. Find the critical height of the slope.**

There are four main types of failures which are:

1. Translational failure
2. Rotational failure
3. Wedge Failure
4. Compound failure

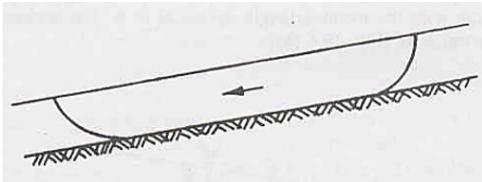


Fig 5: Rotational Failure

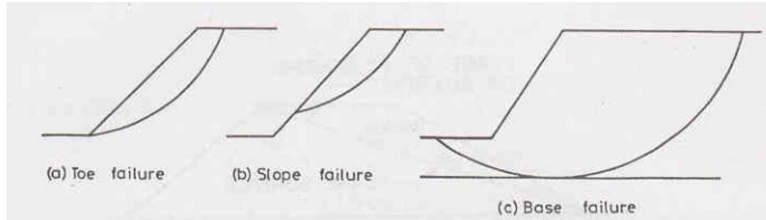


Fig 5: Rotational Failures and it's types

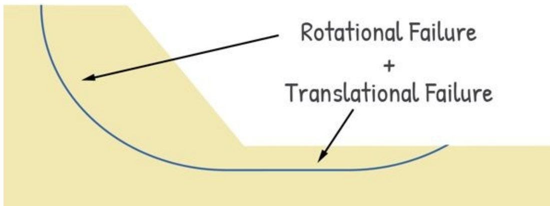


Fig 5: Compound Failure

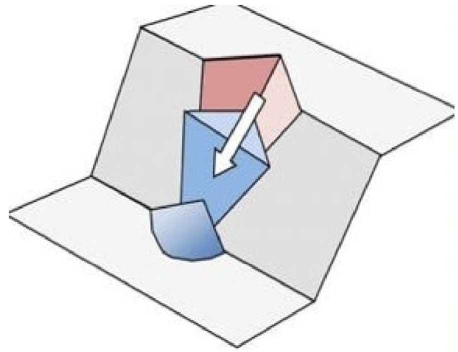


Fig 5: Wedge Failure

Numerical Portion:
Given that;
Taylor's stability number (S_n) = 0.261
Unit weight of soil (γ) = 16 kN/m³
Clay deposit. So, $\varphi = 0$
Unit ultimate cohesion (C) = 30 kN/m²
Factor of safety with respect to cohesion (F_n) = 1
From Taylor's stability number,

$$S_n = \frac{C}{F_c \gamma H}$$

Height of slope,

$$H = \frac{C}{S_n F_c \gamma}$$

$$= \frac{30}{0.261 \times 1 \times 16}$$

$$= \frac{30}{4.176}$$

$$= 7.18 \text{ m}$$

Also, $\left(F_c = \frac{H_c}{H}\right)$
Critical height of slope (Hc) = Fc × H = 1 × 7.18 m
Thus, a vertical cut is made in a clay deposit upto 7.18 m.

Section B

5. **What effect do you foresee on hydrological cycles due to climate change? Suggest the ways and means in mitigating climate change effects on irrigation projects. [5]**

Effects of Climate Change on the Hydrological Cycle:

a) Change in Precipitation Pattern
→ Unpredictable and uneven rainfall distribution
→ Increase in intense rainfall events causing floods
b) Glacier and Snow Melt
→ Accelerated melting of glaciers and snow in Himalayan region
→ Risk of Glacial Lake Outburst Floods (GLOFs)
c) Altered River Flow Regime: Higher monsoon flows and reduced dry-season flows
d) Increased Evapotranspiration: Rising temperature increases evaporation from reservoirs, canals and soil
e) Groundwater Impact: Over-extraction leading to declining water table
f) Increased Frequency of Extreme Events
→ Floods damaging irrigation infrastructure
→ Droughts causing crop failure and water stress

Mitigation Measures for Climate Change Effects on Irrigation Projects

a) Climate-Resilient Planning and Design: Design irrigation systems considering future climate scenarios

b) Efficient Water Management
→ Adoption of water-saving irrigation methods (drip, sprinkler)
→ Lining of canals to reduce seepage losses
→ Proper water scheduling and rotation

c) Storage and Rainwater Harvesting
→ Construction of reservoirs, ponds and check dams
→ Promotion of rainwater harvesting to store excess monsoon water

d) Watershed and Catchment Management
→ Afforestation and soil conservation measures
→ Control of erosion and sedimentation in canals and reservoirs

e) Groundwater Recharge Measures
→ Recharge ponds and infiltration trenches
→ Conjunctive use of surface and groundwater

f) Crop and Agricultural Practices
→ Promotion of drought-resistant and low-water-demand crops
→ Adjustment of cropping calendar according to rainfall pattern

g) Institutional and Policy Measures
→ Strengthening irrigation management institutions
→ Capacity building of farmers
→ Integration of climate adaptation strategies in irrigation planning

6. **What is field investigation for a hydropower project? Differentiate between geological and geotechnical investigations. Explain the step-by-step procedure to estimate probabilistic seismic intensity for target probability of exceedance in specified years at a hydropower project site. [10]**

Field Investigation for a Hydropower Project

Field investigation refers to systematic collection of site-specific data required for planning, design and construction of hydropower projects.

Components
→ Topographical survey
→ Geological investigation
→ Geotechnical investigation

→ Hydrological and meteorological study
→ Seismic and tectonic investigation

Difference Between Geological and Geotechnical Investigations

Aspect	Geological Investigation	Geotechnical Investigation
Purpose	Understand regional and site geology	Determine engineering behavior of soil and rock
Focus	Rock type, structure, faults, folds	Strength, deformability, permeability
Scale	Regional and site level	Site-specific and structure-specific
Methods	Geological mapping, aerial photos	Boreholes, SPT, core logging
Output	Geological model	Design parameters
Use	Site selection and layout	Foundation and structural design

To estimate probabilistic seismic intensity for a hydropower site, you should focus on these four core stages of the **Probabilistic Seismic Hazard Analysis (PSHA)**:

- **Seismic Source Identification:** Define all potential earthquake sources (faults and area zones) within a 200–300 km radius of the hydropower site. This involves compiling a historical earthquake catalog and mapping geological structures.
- **Recurrence Characterization:** For each source, calculate the frequency of different earthquake magnitudes using the **Gutenberg-Richter law** ($\log_{10} N = a - bm$). This determines how often small versus large earthquakes are expected to occur.
- **Ground Motion Prediction:** Select appropriate **Ground Motion Prediction Equations (GMPEs)** to estimate the intensity (like PGA) at the site. These models account for the magnitude, the distance from the source, and the specific soil/rock conditions of the dam site.
- **Probability Integration (Hazard Curve):** Combine the source data and attenuation models using the total probability theorem. The final intensity is selected based on the target **Probability of Exceedance** (e.g., 10% in 50 years) using the Poisson distribution:

$$P(Z > z) = 1 - e^{-\lambda_z T}$$

7. Explain various methods of reclamation of waterlogged areas? Explain the specific consideration to be provided for the design, operation and management of Hill Irrigation system. [4+6]

Methods of Reclamation of Water-Logged Land:

→ **Surface Drainage:**
Removal of excess surface water through open drains and field ditches.
→ **Sub-Surface Drainage:**
Installation of tile or perforated pipe drains to lower groundwater table.
→ **Vertical Drainage:**
Pumping of groundwater using tube wells to control water table rise.
→ **Canal Lining & Seepage Control:**
Lining of canals and watercourses to reduce seepage losses causing water logging.
→ **Improved Irrigation Practices / Bio-Drainage:**
Controlled irrigation, efficient methods (sprinkler/drip) and plantation of high water-consuming trees.

The considerations for designing, operating, and managing a hill irrigation system can be categorized into:

1. Technical Design Considerations

Intake Location: should be positioned where springs or streams are naturally running downward.
Canal Alignment: should be aligned through **ridges** to reduce the costs associated with lining and minimize the need for complex cross-drainage works.

2. Social and Managerial Considerations

Success in hill irrigation depends heavily on local community dynamics:

Social Arrangements: The system must respect local traditions, ethnic castes, and village boundaries. It is crucial to account for existing water rights.
Operational Simplicity: If the local farmers lack advanced managerial or institutional capabilities, the irrigation system must be designed to be simple to operate.
Farmer Participation: Farmers should be involved in the engineering design phase. Their "intimate familiarity" with the land provides engineers with vital data on:

- Design flood discharge and river flood levels.
- Movement of boulders and sediments in the river.

3. Agricultural and Financial Factors

Agricultural Needs: The design must be tailored to the specific **type of soil** and the **types of crops** being cultivated in the region.
Financial Constraints: Limitations in budget directly influence the choice of technology, construction methods, and the type of materials used for the project.

Section C

8. What are the basic factors which influence the visible dimension of a road? What correction in required in horizontal curves in negotiating the speed and stability of vehicles? [10]

The visible dimensions and geometric design of a road are influenced by several basic factors:

a) Design Speed

Design speed is the "safe maximum speed" adopted for a road. According to the **NRS-2070** standards, speed recommendations vary based on:

Terrain Type: Plain, Rolling, Mountainous, or Steep.

Road Class: Class-I through Class-IV.

b) 2. Design Vehicle Characteristics

The physical dimensions of the vehicles intended to use the road.

Length of vehicle: (e.g., 18m for certain standards).

Width of vehicle: (Standardized at 2.5m).

Maximum height: (Standardized at 4.75m).

Wheelbase length: (Influences the widening of roads on curves).

c) Topography and Physical Features

The natural layout of the land significantly impacts where and how a road is built:

→ **Hilly Regions:** Dimensions are heavily affected by hills, valleys, steepness of slopes, and stream crossings.

→ **Plain Areas:** Influenced more by drainage requirements and grade separations.

→ **Man-made Features:** Existing structures or infrastructure that the road must navigate around.

d) Traffic Volume and Capacity

The amount and type of traffic determine the scale of the road:

→ **Traffic Volume:** The number of vehicles passing a section in a specific time.

→ **Traffic Capacity:** The number of vehicles a lane can handle per unit of time. This determines the number of lanes required (e.g., the formula $C = \frac{1000 \times V}{S}$ is used to calculate capacity).

→ **30th Highest Hourly Volume:** This specific metric is used to design highway facilities to ensure they aren't under-designed for peak times.

e) Road User Characteristics
The human element—including the education, awareness, and "civic traffic sense" of the people using the road—influences design.

f) Environmental Factors
Aesthetics, noise pollution, air pollution, and landscaping are considered to ensure the road integrates safely and pleasantly into its surroundings.

Second Part:
To ensure the stability and safety of a vehicle negotiating a horizontal curve, two primary corrections are required: **Superelevation** and **Widening of Pavement**.
A. Superelevation (Cant)
When a vehicle moves around a horizontal curve, centrifugal force acts outwards, which can cause the vehicle to skid or overturn. To counteract this, the outer edge of the road is raised above the inner edge.
The general formula used for design is:

$$e + f = \frac{V^2}{127R}$$

e: Rate of superelevation.
f: Coefficient of lateral friction (usually taken as **0.15**).
V: Design speed (taken from the tables in your image, e.g., **100 kmph** for Class-I rolling terrain).
R: Radius of the curve.

B. Extra Widening on Curves
When a long vehicle negotiates a curve, the rear wheels do not follow the same path as the front wheels (off-tracking). To accommodate this, the pavement is widened. This widening consists of two parts:
Mechanical Widening (W_m): Required due to the rigidity of the wheelbase.

$$W_m = \frac{nl^2}{2R}$$

(Where n is the number of lanes and l is the wheelbase length).
Psychological Widening (W_{ps}): Required to provide more space for drivers who tend to maintain a greater clearance from the pavement edge at higher speeds.

$$W_{ps} = \frac{V}{9.5\sqrt{R}}$$

C. Transition Curves
To prevent a sudden change in centrifugal force (which would be uncomfortable and unstable), a transition curve is introduced between the straight path and the circular curve. This allows for:
→ Gradual introduction of **superelevation**.
→ Gradual introduction of **extra widening**.

9. What are the general requirements of traffic control devices? What are traffic islands? Briefly describe various types of traffics islands.
General Requirements for Traffic Control:

→ **Minimize Conflict Points:** Reducing the number of spots where vehicle paths cross.
→ **Small Approach Speed:** Encouraging lower speeds as vehicles enter the area.
→ **Adequate Visibility:** Ensuring drivers have a clear line of sight for other vehicles.
→ **Proper Signage:** Clear and visible signs must be provided.
→ **Good Lighting:** Essential for safety during nighttime driving.
→ **Avoid Sudden Changes:** Preventing abrupt path changes that could lead to accidents.
→ **Geometric Adequacy:** Providing an adequate turning radius and pavement width.

Traffic Islands

A **Traffic Island** is a raised area constructed within a roadway. Its primary purpose is to avoid or minimize areas of major or minor conflict between vehicles or between vehicles and pedestrians.

Types of Traffic Islands

I. Divisional Island

These are placed in the middle of the road to separate opposing flows of traffic. Eliminates head-on collisions and reduces headlight glare during the night.

II. Channelizing Island

Located specifically at intersections to guide traffic into the correct lanes. It streamlines traffic flow and reduces driver confusion at complex junctions.

III. Refuge Island

Often used in conjunction with divisional islands at crosswalks. To provide a safe place for pedestrians to stop midway while crossing a wide street.

IV. Rotary Island

A large central island provided at the intersection of multiple roads.

a) It forces all approaching traffic to move in a clockwise direction and then "weave out" toward their respective destinations.
b) It converts dangerous crossing maneuvers into safer **weaving maneuvers**, significantly minimizing conflict points.

10. List the sequence of layers and test requirements in the field to construct a flexible pavement of parallel taxiway. [2+3=5]

A **parallel taxiway** in an airport requires a **flexible pavement** capable of withstanding repeated aircraft wheel loads. As per **ICAO guidelines**, **CAAN practice**, and **IRC standards** adopted in Nepal, the pavement is constructed in the following sequence from bottom to top.

Subgrade → GSB → Base → Binder (DBM) → Wearing (BC)

a) Prepared Subgrade (The Foundation)

→ Compacted natural soil or select fill.
→ **Key Field Tests:**
Field Density Test: Sand Replacement method or core cutter
CBR Test: Laboratory on remoulded or soaked samples

b) Sub-base Course (Granular Sub-base - GSB)

→ Crushed aggregates or natural gravel for drainage and load spread.
→ **Key Field Tests:**
Gradation test of aggregates

Atterberg limits (for fines)
Los Angeles abrasion test

c) Base Course (WMM / WBM)

→ Wet Mix Macadam (WMM) is preferred for airports due to better interlocking.

→ **Key Field Tests:**

Aggregate crushing value / impact value
Gradation test
Field density test
CBR or modulus test

d) Binder Course (DBM)

→ Dense Bituminous Macadam (DBM) acting as the main structural bituminous layer.

→ **Key Field Tests:**

Bitumen penetration / viscosity test
Marshall stability and flow test
Bitumen content test
Degree of compaction
Thickness and level check

e) Surface/Wearing Course (Bituminous Concrete - BC)

→ The topmost layer providing skid resistance and waterproofing.

→ **Key Field Tests:**

Core Test: To verify field density and thickness.
Surface Evenness: Using a 3-meter straight edge.
Skid Resistance: To ensure aircraft braking safety.

Section D

11. What are the major factors governing water demand and variation of water demand? Briefly describe in context of Nepal. [10]

Major Factors Governing Water Demand

→ **Size and Type of Community:** Larger cities generally have higher per capita demand.

→ **Standard of Living:** People with higher economic status tend to use more water for luxuries like car washing, gardening, and flushing toilets.

→ **Climatic Conditions:** Hot and dry climates see significantly higher demand for drinking, bathing, and irrigation compared to cool or humid climates.

→ **System of Supply:**

- **Continuous Supply:** Provides water 24/7, often leading to higher consumption but more stable pressure.
- **Intermittent Supply:** Common in Nepal, where water is only available for a few hours. This often leads to lower overall consumption but creates extreme "rush hour" peaks.

→ **Quality and Pressure of Water:** If the water is clean and safe, consumption increases.

→ **Industrial and Commercial Presence:** Areas with many hotels, hospitals, or factories require much larger volumes of water.

Variation of Water Demand

Water demand is never constant; it fluctuates throughout the year, the week, and the day. Engineers use **Peak Factors** to design components to handle these maximum loads.

→ **Seasonal Variation:** Demand is highest in the dry summer months and lowest in the rainy season.
→ **Daily Variation:** Consumption varies by day of the week (higher on holidays or festivals like Dashain and Tihar).
→ **Hourly Variation:** Demand typically peaks in the early morning (cooking and cleaning) and evening, while staying near zero during the night.

Peak Factors in Nepal

Engineering designs in Nepal often adopt specific multipliers to ensure pipes don't run dry during peak hours:

→ **Seasonal Peak Factor:** Often taken as **1.0** (or ignored) because seasonal variation is relatively low compared to daily fluctuations.
→ **Daily Peak Factor:** Generally, **1.1 to 1.5** (Max daily demand = 1.5 x Avg daily demand).
→ **Hourly Peak Factor:** Usually, **3.0** for urban systems, meaning the network must handle three times the average hourly flow during peak morning hours.

12. **List in detail the physical biological and socio-economic baseline information that have to be collected during EIA study of a water supply project. [10]**

Environmental Impact Assessment (EIA) is a mandatory tool for environmentally significant water supply projects in Nepal as per the **Environment Protection Act, 2019** and **Environment Protection Rules, 2020**.

Baseline information represents the **existing physical, biological and socio-economic condition** of the project area prior to project implementation.

1. Physical Environment Baseline

- **Topography and Geology:** Landforms, slope stability, and soil types along the pipeline alignment and intake sites.
- **Hydrology:** Flow measurements of the source river/spring (dry season flow vs. peak flow).
- **Water Quality:** Baseline testing of physical (turbidity, color), chemical (pH, Iron, Arsenic, Fluoride), and biological (Total Coliform) parameters against National Drinking Water Quality Standards (NDWQS).
- **Climate:** Temperature, precipitation patterns, and humidity of the project area.
- **Air and Noise Quality:** Ambient noise levels at construction sites (e.g., near pump houses or treatment plants) and dust levels.
- **Land Use:** Current land use patterns—forest, agricultural, or residential—that will be converted for project structures.

2. Biological Environment Baseline

This focuses on the flora and fauna that may be disturbed during construction or by water diversion.

- **Flora:**
    - Inventory of tree species in the project area
    - Identification of **protected or endangered species** under Nepali law or CITES.
    - Density and volume of trees to be cleared
- **Fauna:**
    - Mammals, birds, and reptiles found in the project's **Zone of Influence (ZoI)**.
    - **Aquatic Life:** Fish species in the source river, focusing on migratory patterns and breeding grounds that might be affected by the intake weir.
- **Ecosystem Services:** Identification of wetlands or "Sacred Groves" (religious forests) that provide ecological value.

3. Socio-economic and Cultural Baseline

This assesses how the project will impact human lives, livelihoods, and heritage.

- **Demography:** Population size, household number, ethnicity, and gender distribution in the affected Wards.
- **Livelihood and Economy:** Primary occupations (agriculture, trade, etc.), average income levels, and land ownership status.
- **Existing Infrastructure:** Current water supply status, sanitation facilities, proximity to schools, hospitals, and electricity lines.
- **Downstream Water Use:** Documentation of existing irrigation canals (Kulos), water mills (Ghatta), or traditional stone spouts (Dhunge Dhara) that rely on the same source.
- **Public Health:** Prevalent water-borne diseases in the area (e.g., Diarrhea, Typhoid) to establish a "before-project" health benchmark.
- **Cultural and Religious Sites:** Presence of temples, cremation ghats, or archaeological sites near the intake or distribution network.

THE END